Device description

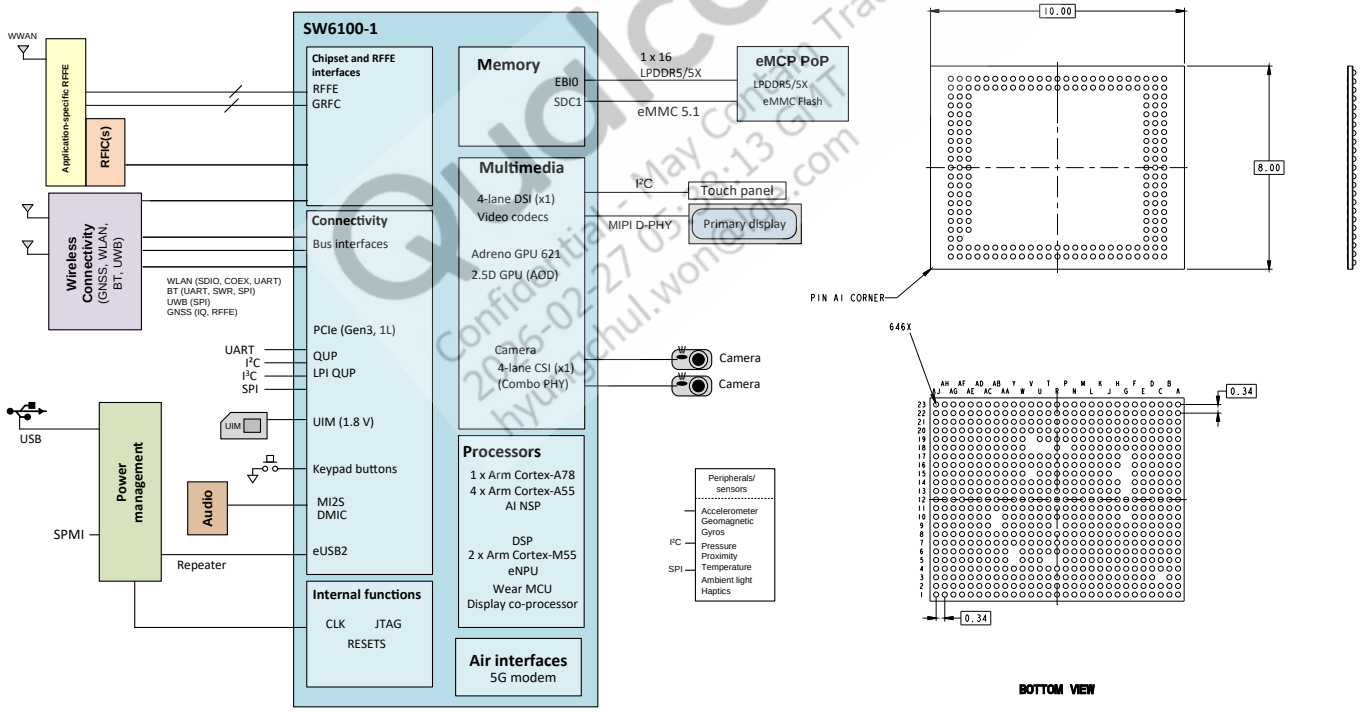
The SW6100-1/SW6100P-1 is a 3 nm process chip with Qualcomm® Snapdragon™ processor supporting 4G/5G sub-6 technologies designed exclusively for wearable designs. Key processor and memory characteristics include:

- 8.0 mm × 10.0 mm x 0.36 mm package-on-package (PoP) with 0.34 mm bottom pitch and 0.35 mm top pitch (PoP)
- Single Arm Cortex-A78 applications processor supporting up to 2.1 GHz, with 256 kB of L2 cache and 32 kB of dedicated L1 cache. Quad Core Arm Cortex-A55 applications processor supporting up to 1.95 GHz, with 64 kB of dedicated L2 cache and 32 kB of dedicated L1 cache per core
- 3 MB of shared L3 cache between all the cores
- Qualcomm® Adreno™ 621 graphics processing unit (GPU). Dedicated 2.5D LPI GPU for always-on display (AoD)
- Integrated wearable microcontroller (WM) system and ambient-mode microcontroller (AM) to handle low power display, audio, and sensor use cases with dual Arm Cortex M55 cores
- Display coprocessor (DCP) for always-on display (AOD) support
- Dedicated digital signal processor (DSP) for Snapdragon Sensor Core and low-power audio subsystem
- Dedicated neural signal processor (NSP) for AI applications
- Dual-embedded neural processing units (e-NPU) for high-performance and always-on neural network (NN) use cases
- Package-on-package (PoP) high-speed eMCP memory with 1 × 16 up to 4.2 GHz LPDDR5/5X SDRAM up to 4 GB and eMMC 5.1 up to 64 GB
- Always-on subsystem with RPM for power management

Key features

- Qualcomm® universal bandwidth compression (UBWC) with GPU
- Embedded deep trench capacitors (DTC) in package, to improve PDN performance and reduce layout complexity
- 720 × 720 60 Hz, 4-layers, 1 × 4-lane DSI (D-PHY v1.2)
- 13 + 13 MP, 25 MP, 1080p60, 1 × 4-lane (D-PHY 1.2/C-PHY 1.2)
- 1 × eUSB2 support, with repeater integrated inside the PMIC
- PCIe Gen3 1-lane
- 1.2 V/1.8 V capable I/Os
- Multiple QUP engines to support I²C/UART/SPI/I³C
- Support for PSRAM with Hexadecimal/Octal SPI interface
- Supports 3GPP Rel. 17 RedCap with multimode 4G and 5G sub-6 SA transceiver SDR125
- QCC6100 chip attach to SoC offering:
  - WLAN IEEE 802.11a/b/g/n/ac/ax with Tri-band (2.4 GHz/5 GHz/ 6 GHz)
  - Bluetooth® v6.0 with Bluetooth low energy (BLE) audio
  - L1 and L5 GNSS support for navigation applications
  - UWB: 1 Tx + 1 Rx configuration

SW6100-1/SW6100P-1 high-level block diagram



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1 Introduction

Document updates

See the Revision history for details on the changes included in this revision.

1.1 Functional block diagram

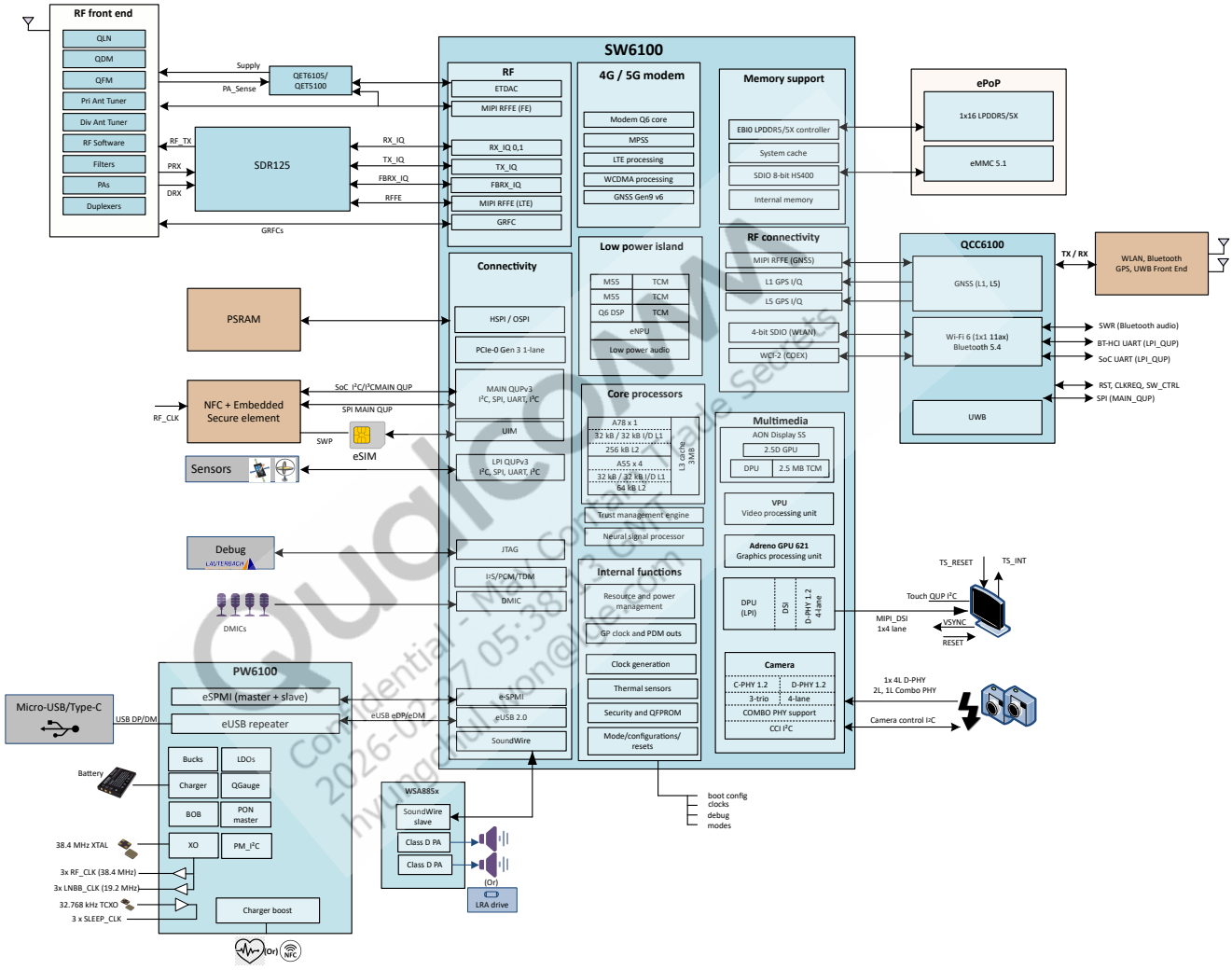

Figure 1-1 SW6100-1 functional block diagram and example application

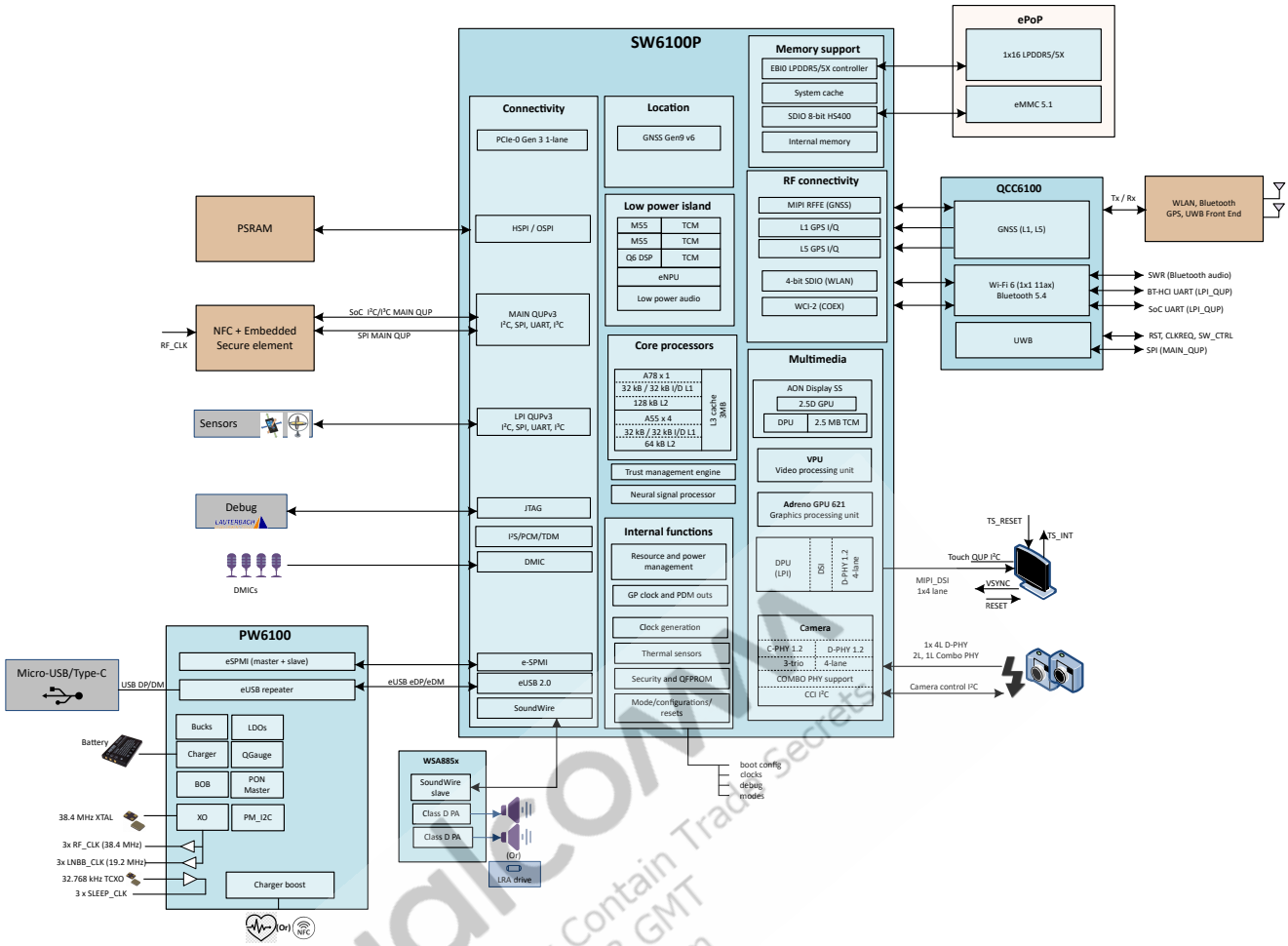

Figure 1-2 SW6100P-1 functional block diagram and example application

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1.2 Device variants

The following table summarizes the SW6100-1/SW6100P-1 device variants.

Device	SKU	Modem	Connectivity	AI core	Satelite	Camera	Peripherals
SW6100-1	AA	5G RedCap/4G/3G	Wi-Fi/Bluetooth/GNSS/UWB	WearAI	NB-NTN	Camera	PCIe
SW6100P-1	AA	No modem	Wi-Fi/Bluetooth/GNSS/UWB	WearAI	Without NB-NTN	Camera	PCIe

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1.3 SW6100-1/SW6100P-1 features

Table 1-1 SW6100-1/SW6100P-1 features

Feature	SW6100-1/SW6100P-1 capability
Processors	
Applications	64-bit applications processor ▪ Arm Cortex-A78 applications processor up to 2.1 GHz, with 256 kB L2 cache ▪ Quad Core Arm Cortex-A55 applications processor up to 1.95 GHz, with 64 kB L2 dedicated cache per core ▪ 3 MB of shared L3 cache between all the cores
Supported OS	Android, Wear OS
Always-on system	Always-on subsystem with modem power management unit
Co-processor	
Low-power audio island and sensors	▪ Q6 V79 digital signal processor (DSP) ▪ Wear microcontroller (WM) and ambient mode microcontroller (AM) subsystem each equipped with Arm Cortex-M55 low power ▪ Dual-embedded neural processing units (e-NPU) for high-performance and always-on neural network (NN) use cases ▪ Low-power audio island (LPAI) to handle audio use cases ▪ Snapdragon Sensor Core (SSC) for low power sensor use cases
Memory support	
EBI	▪ Package on package (PoP) ▪ 1 × 16-bit LPDDR5/5X with 3200/4266 MHz up to 4GB ▪ 128 kB LLC dedicated for DARE, MACH9
eMMC	eMMC 5.1 up to 64 GB
PSRAM	OSPI/HSPI with 266 MHz DDR
RF support	
RF operating bands	▪ Defined by the RF transceiver SDR125 ▪ 5G NR sub-6 (FR1) SA and multimode 4G LTE
CA support	Not supported
3GPP release	3GPP Rel. 17 with 5G reduced capability (RedCap) support
GRFC and RFFE interfaces	13 GRFC and 7 RFFE interfaces
WLAN	▪ IEEE 802.11a/b/g/n/ac/ax [a] ▪ Tri-band 2 GHz/5 GHz/6 GHz [a] ▪ 1 × 1 2.4 GHz/5 GHz/6 GHz 20 MHz with 11ac [a] ▪ 1 × 1 2.4 GHz/5 GHz/6 GHz 20 MHz with 11ax [a]
Bluetooth	▪ Compliant with Bluetooth v6.0 support [a] ▪ SoundWire (SWR) interface for Bluetooth low energy (BLE) audio [a]
UWB	▪ 1Tx + 1Rx configuration [a]
GNSS	▪ Concurrent support for L1 and L5 bands [a] ▪ Dedicated I/Q interfaces for communication [a]

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Table 1-1 SW6100-1/SW6100P-1 features (cont.)

Feature	SW6100-1/SW6100P-1 capability
Multimedia	
Display support	
Mobile display processor	Adreno DPU 1122
MIPI DSI	1 × DSI 4 lanes (D-PHY 1.2)
General display features	▪ 720 × 720 60 fps, 4-layers ▪ Supports UBWC4
Wear display subsystem (WDSS)	▪ Display control processor (DCP) with 2.25 MB TCM [b] ▪ 2.5D GPU with 3.75 MB TCM
Camera support	
Camera interfaces	13MP + 13MP (dual), 25 MP (single) with 1080p60
MIPI CSI	▪ 1 × CSI D-PHY with 4-lane (1.2)/C-PHY (1.2) with 3-Trio ▪ Supports combo-mode (C-PHY + D-PHY) on the same port
Video applications performance	
Encode	AR50LT 1080p60e
Decode	AR50LT 1080p60d
Concurrency	Supports concurrent 1080p60e + 1080p60d
Graphics performance	
Graphics	Adreno A621 GPU with: ▪ 701 MHz nominal ▪ 843 MHz turbo ▪ 3D graphics accelerator with 64-bit addressing
Audio	
Codec/Amplifier	Through WSA885x attach
Attaches	▪ PMW6100-0: Only PMIC, no amplifier/haptics
Voice UI	▪ Supports 2 voice activation engines ▪ Supports always-on noise suppression
Low-power voice activation	Supported
Audio interfaces	▪ 3 × MI^{2}S interface with 2-data lanes each ▪ 3 × DMIC interface with CLK, DATA ▪ 2 × 2-lane SWR (SoundWire) for the dual WSA amplifiers ▪ 1 × 2-lane SWR (SoundWire) for QCC6100 BLE audio ▪ 1 × 6-lane SWR (3-Tx, 1-Rx, CLK, DATA) Some features may not be simultaneously available due to multiplexing. For more information, see *SW6100/SW6100P Pin Assignment and GPIO Configuration Spreadsheet*.
Voice codec	Fluence AI, 32 kHz EVS, Stereo-MIC, Speaker protection alert
Voice processing	Qualcomm® Noise and Echo Cancellation and Qualcomm® Voice Suite
Audio codec support	High resolution 192 kHz audio, AAC family (HE-AAC, HEAAC v2), WMA family (WMA and WMA Pro+)
Enhanced audio	DSP-offload for low-power audio playback
Connectivity	

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Table 1-1 SW6100-1/SW6100P-1 features (cont.)

Feature	SW6100-1/SW6100P-1 capability
Qualcomm universal peripheral (QUP) ports	■ 11 serial engines on TLMM Main QUP ■ 14 serial engines on LPASS QUP
CCI I2C	2 dedicated I²C interfaces for the camera: ■ CCI0_I2C: 1.2 V only ■ CCI1_I2C: 1.2 V/1.8 V
UIM	1 × 1.8 V UIM interface (supports eSIM and Class-C nano-SIM)
eUSB2	Embedded USB2 (eUSB2) Standard support with repeater integrated inside PW6100 attach
PWM	10 GPIOs support PWM functionality
Secure digital interfaces	■ 1 × 8-bit 1.8 V only SD 3.0 for eMMC 5.1 over PoP ■ 1 × 4-bit 1.2 V only SD 3.0 for interface with QCC6100 for WLAN (multiplexed with GPIO)
Touchscreen support	Capacitive panels via external IC (I^2C and interrupts)
PCIe	1 × PCIe Gen3, 1-Lane
PSRAM	For PSRAM support, 266 MHz DDR over OSPI/HSPI
Configurable GPIOs	
Number of GPIO ports	■ 150 overall with 13 (LPASS), 44 (SSC) ■ Some features may not be simultaneously available due to multiplexing. For more information, see *SW6100/SW6100P Pin Assignment and GPIO Configuration Spreadsheet*.
Top-level mode multiplexer	Provides a convenient way to program groups of GPIOs
Internal functions	
Security	TME 2.2, ICE 4.0, LCP-DARE, QTEE5.x, QRNG, QFPROM
PLLs and clocks	■ Multiple clock regimes; watchdog and sleep timers input: 19.2 MHz CXO ■ General-purpose outputs
Debug	JTAG, embedded USB debug (EUD), and ETM
Others	Thermal sensors, modes and resets, and peripheral subsystem
Fabrication technology and package	
PoP, thermally efficient package	8.0 mm × 10.0 mm x 0.36 mm package-on-package (PoP)

[a] Provided through QCC6100 chip attach.
[b] For always-ON (AON) use cases.

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2 Pin definitions

2.1 I/O parameter definitions

Table 2-1 I/O description (pin type) parameters

Symbol	Description
Pin attribute	
AI	Analog input (does not include pin circuitry)
AO	Analog output (does not include pin circuitry)
B	Bidirectional digital with CMOS input
DI	Digital input (CMOS)
DO	Digital output (CMOS)
H	High-voltage tolerant
S	Schmitt trigger input
Z	High-impedance (Hi-Z) output
Pin pull details for digital I/Os	
nppdpukp	Programmable pull resistor. The default pull direction is indicated using capital letters, and is a prefix to other programmable options: ▪ NP: pdpukp = default no-pull, with programmable options following the colon (:) ▪ PD: nppukp = default pull-down, with programmable options following the colon (:) ▪ PU: nppdkp = default pull-up, with programmable options following the colon (:) ▪ KP: nppdpu = default keeper, with programmable options following the colon (:)
KP	Contains an internal weak keeper device (keepers cannot drive external buses)
NP	Contains no internal pull
PU	Contains an internal pull-up device
PD	Contains an internal pull-down device
Pin-voltage groupings for baseband circuits	
PX0	Pin group 0 (Deep sleep, Reset, SPMI)
PX1	Pin group 1 (DDR RESET)
PX3	Pin group 3 (Reset Out)
PX11	Pin group 11 (CXO)
PX14	Pin group 14 (PS_HOLD, sleep clock, some GPIOs)
PX701	Pin group (I/O pad) 701: 1.2 V/1.8 V
PX702	Pin group (I/O pad) 702: 1.2 V/1.8 V
PX703	Pin group (I/O pad) 703: 1.2 V/1.8 V
PX704	Pin group (I/O pad) 704: 1.2 V/1.8 V
PX705	Pin group (I/O pad) 705: 1.2 V/1.8 V
PX706	Pin group (I/O pad) 706: 1.2 V/1.8 V
PX707	Pin group (I/O pad) 707: 1.2 V/1.8 V
PX709	Pin group (I/O pad) 709: 1.2 V/1.8 V

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Table 2-1 I/O description (pin type) parameters (cont.)

Symbol	Description
PX711	Pin group (I/O pad) 711: 1.2 V/1.8 V
PX712	Pin group (I/O pad) 712: 1.2 V/1.8 V
PX714	Pin group (I/O pad) 714: 1.2 V/1.8 V
PX715	Pin group (I/O pad) 715: 1.2 V/1.8 V

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2.2 Pin assignments – bottom

2.2.1 Pin map – bottom

The SW6100-1/SW6100P-1 is available in the FOPPSP646. Its bottom surface is equivalent to a FOPPSP646 that includes several ground pins for electrical grounding, mechanical strength, and thermal continuity. See Mechanical information and Pin assignments – top for information about the top pin assignments. A high-level view of the bottom pin assignments is shown in the following figure.

The text within the following figure is difficult to read when viewing an 8½ inch × 11 inch hard copy. Other viewing options are available:

- Print that one page on an 11 inch × 17 inch sheet.
- View the graphic's PDF soft copy and zoom in — the resolution is sufficient for comfortable reading.
- Download the *SW6100/SW6100P Pin Assignment and GPIO Configuration Spreadsheet (80-83950-1A)*. This Microsoft Excel spreadsheet lists all pin numbers (in alphanumeric order), pin names, pin voltages, pin types, and functional descriptions.

NOTE Click the following link to download the pin assignment and GPIO configuration spreadsheet.

SW6100/SW6100P Pin Assignment and GPIO Configuration Spreadsheet

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	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
A	GND	GPIO_123	GPIO_15	GPIO_14	GPIO_72	GPIO_86	GPIO_82	DNC	DNC	GND	GND	GND	GND	GND	GND	GND	BBRX1_I_P	GND	VCCI_EMMC	GND	VCCQ_EMMC	VCC_EMMC	GND
B	GPIO_121	GPIO_124	GPIO_102	GPIO_40	GPIO_97	GPIO_84	GPIO_80	RESOUT_N	GND	GNSS_L5_I	GND	TXDAC_Q_M	GND	TXDAC_I_P	GND	BBRX0_Q_P	BBRX1_I_M	GND	GND	GPIO_49	GPIO_54	GPIO_59	GPIO_56
C	GPIO_111	GPIO_122		GPIO_96	GND	GPIO_85	GPIO_81	GPIO_83	GND	GNSS_L5_Q	GND	TXDAC_Q_P	GND	TXDAC_I_M	GND	BBRX0_Q_M	GND	GND	FBRX_I	GND	GPIO_55	GPIO_58	GPIO_57
D	GPIO_116	GPIO_112	GPIO_98	GPIO_94	GPIO_44	GPIO_43	SLEEP_CLK	GND	GND	GNSS_L1_Q	GND	GND	GND	TXDAC_REF	BBRX0_I_P	GND	BBRX1_Q_P	GND	FBRX_Q	GND	GPIO_61	GND	GPIO_63
E	GPIO_115	GPIO_99	GPIO_95	GPIO_41	GPIO_42	GPIO_117	GPIO_78	MODE_1	GND	GNSS_L1_I	GND	ETDAC_P	ETDAC_M	GND	BBRX0_I_M	GND	BBRX1_Q_M	GND	GND	GPIO_50	GPIO_60	GPIO_67	GPIO_62
F	GPIO_106	GND	GPIO_126	GPIO_100	GPIO_118	GPIO_29	GPIO_149	MODE_0	GND	GND	GND	GND	GND	GND	GND	VDD_AL_BBRX	VDD_AL_FBRX	GND	GPIO_51	GPIO_52	GPIO_66	GPIO_64	GPIO_65
G	GPIO_114	GPIO_105	GPIO_125	GPIO_101	GPIO_28	GND	VDD_LPI_CX	VDD_PX701	PS_HOLD		VDD_A_GNSS_ADC_0P9	VDD_MX_A_TXDAC_ETDAC						VDD_GFX	GND	GPIO_53	GPIO_147	GPIO_45	GPIO_46
H	DNC	GPIO_104	GPIO_103	VDD_PX703	VDD_PX14	GND	VDD_LPI_CX	VDD_LPI_MX	VDD_PX3	VDD_LPI_CX	GND	VDD_MX_A	VDD_AH_TXDAC_ETDAC	GND		VDD_GFX	GND	VDD_GFX	GND	GPIO_47	GPIO_48	GPIO_148	GPIO_16
J	REFGEN_REXT0	GPIO_107	GPIO_109	GPIO_119	VDD_LPI_CX	GND	VDD_LPI_CX	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	VDD_GFX	GND	VDD_GFX	VDD_PX705	GPIO_19	GPIO_17	GPIO_18	GND
K	GND	GND	GPIO_113	GPIO_110	GPIO_120	VDD_PX704	VDD_LPI_CX	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	VDD_GFX	GND	VDD_GFX	VDD_PX706	GND	GPIO_79	GPIO_68	GPIO_32
L	PCIE0_RX0_P	PCIE0_REFCLK_P	GND	GPIO_108	GPIO_93	GND	VDD_LPI_CX	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	VDD_GFX	GND	VDD_GFX	VDD_PX14	GPIO_70	GPIO_71	GPIO_69	GPIO_34
M	PCIE0_REFCLK_M	PCIE0_RX0_M	GND	GPIO_92	VDD_IX	VDD_PX14	VDD_LPI_CX	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	VDD_CX	GND	VDD_QFPROM	VDD_IX	GPIO_33	GPIO_38	GPIO_35	GPIO_37
N	LPDDR5_VDD2H	GND	GPIO_90	GPIO_91	VDD_PX702	DNC	GND	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	VDD_CX	GND	SPMI0_CLK	VDD_PX709	JTAG_TCK	JTAG_TDI	GPIO_39	GPIO_36
P	PCIE0_TX0_M	PCIE0_TX0_P	GND	GND	VDD_PX11	CXO	VDD_A_REFGEN_0P875_0	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	VDD_CX	SPMI0_DATA		GND	JTAG_TRST_N	JTAG_TMS	JTAG_TDO	JTAG_SRST_N
R	LPDDR5_VDD2L	GND	EUSB0_EDM	EUSB0_EDP		VDD_A_PCIE_0_0P9	VDD_A_REFGEN_1P2_0	VDD_LPI_MX		VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	GND	VDD_PX0	GND	VDD_PX707	DNC	GND	REFGEN_REXT1	GND
T	DSI0_LN3_P	DSI0_LN3_M	GND	VDD_A_DSI_0_0P9		VDD_A_EUSB_0_0P9	GND	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	GND	VDD_LPI_MX		VDD_PX711	GND	CSI0_A0_CLK_M	CSI0_NC_CLK_P	LPDDR5_VDD2L
U	GND	DSI0_LN2_M	DSI0_LN2_P	GND	VDD_A_DSI_0_PLL_0P9	VDD_A_PCIE_0_1P2	VDD_A_EUSB_0_1P2	VDD_LPI_MX	GND	VDD_LPI_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	GND	VDD_A_REFGEN_1P2_1		VDD_A_REFGEN_0P875_1	CSI0_B0_LN0_P	CSI0_C0_LN0_M	GND	LPDDR5_VDD2H
V	DSI0_CLK_P	DSI0_CLK_M	DSI0_LN0_P	DSI0_LN0_M	GND	VDD_A_DSI_0_1P2	GND	VDD_LPI_MX	GND	VDD_CX	GND	VDD_MX_A	VDD_CX	GND		GND	VDD_NSP			GND	CSI0_A1_LN1_P	CSI0_B1_LN1_M	GND
W	GND	DSI0_LN1_M	DSI0_LN1_P	VDD_IX	VDD_PX715	GND	VDD_APC0	VDD_APC0	GND	VDD_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	GND	VDD_NSP	VDD_A_CSI_0_0P9	VDD_A_CSI_0_1P2	CSI0_C1_LN2_P	CSI0_A2_LN2_M	GND	LPDDR5_VDD1
Y	LPDDR5_VDD1	GPIO_146	GND	GPIO_145			VDD_MX_C	VDD_APC0	GND	VDD_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	GND	VDD_NSP	VDD_NSP	VDD_PX14	GND	CSI0_C2_LN3_M	CSI0_B2_LN3_P	GND
AA	LPDDR5_VDDQ	GPIO_144	GPIO_143	VDD_PX14	VDD_APC0	GND	VDD_MX_C	VDD_APC0	GND	VDD_CX	GND	VDD_MX_A	VDD_CX	GND	VDD_MX_C	GND	VDD_NSP	GND	VDD_PX712	GPIO_20	GPIO_21	GPIO_22	LPDDR5_VDDQ
AB	GND	GPIO_141	GPIO_142	VDD_PX713	VDD_APC0	GND	VDD_MX_C	VDD_APC0			GND	VDD_MX_A	GND	VDD_NSP	VDD_MX_C	GND	VDD_NSP	GND	VDD_IX	GPIO_23	GPIO_1	GPIO_0	GND
AC	GND	GPIO_140	GPIO_138	VDD_PX0	VDD_APC0	GND	VDD_MX_C	VDD_APC1	GND	VDD_APC1	VDD_A_EBI_0	VDD_A_EBI_0	GND	VDD_A_EBI_0	VDD_A_EBI_0	GND	VDD_NSP	GND	VDD_NSP	GPIO_4	GPIO_6	GPIO_2	GND
AD	GND	GPIO_139	GPIO_136	VDD_PX1	VDD_APC0	GND	VDD_MX_C	VDD_APC1	GND	VDD_APC1	GND	VDD_A_EBI_0_PLL_0P9	VDD_D_EBI_0	GND	GND	GND	VDD_NSP	GND	VDD_NSP	GPIO_5	GPIO_8	GPIO_3	GND
AE	GND	GPIO_137	GPIO_135	GPIO_130	VDD_APC0	GND	VDD_MX_C	VDD_APC1	GND	GND	VDD_IO_EBI_0	VDD_IO_EBI_0	VDD_IO_EBI_0	VDD_IO_EBI_0	GND	GND	VDD_NSP	GPIO_87	GPIO_76	GPIO_11	GPIO_10	GPIO_7	GND
AF	GND	GPIO_134	GPIO_133	GPIO_127	VDD_APC0	VDD_MX_C	GND	VDD_MX_C	VDD_APC1	GND	GND	VDD_IO_EBI_0	VDD_A_EBI_0	VDD_PX1	GND	VDD_MX_C	VDD_NSP	GPIO_77	GPIO_31	GPIO_9	GPIO_25	GPIO_24	GND
AG	GND	GPIO_132	GPIO_129	DS_EN	GND	GND	GND	GND	GND	GND	GND	VDD_IO_EBI_0	VDD_A_EBI_0	GND	GND	GND	GND	GPIO_89	GPIO_74	GPIO_27	GPIO_30	GPIO_26	GND
AH	GND	GPIO_131	GPIO_128	RESIN_N	VDD_APC0	VDD_MX_C	GND	VDD_MX_C	VDD_APC1	GND	GND	GND	GND	GND	GND	VDD_MX_C	VDD_NSP	GPIO_88	GPIO_75	GPIO_73	GPIO_12	GPIO_13	ZQ0
AJ	GND	GND	LPDDR5_VDD2H	LPDDR5_VDD1	GND	LPDDR5_VDDQ	GND	GND	GND	LPDDR5_VDD2H	DNC	LPDDR5_VDD2L	EBI0_CAL	LPDDR5_VDD2H	GND	GND	GND	LPDDR5_VDDQ	GND	GND	LPDDR5_VDD1	LPDDR5_VDD2H	GND

PWR
GND
PCIE
USB
CSI
DSI
EBI
DNC
Other

Board View

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Figure 2-1 SW6100-1/SW6100P-1 bottom pin assignments

2.2.2 Pin descriptions – bottom

The pins are described in Table 2-2 to Table 2-4.

Table 2-2 SW6100-1/SW6100P-1 bottom pin descriptions – general pins

Pin no.	Pin name	Pin characteristics		Functional description
		Voltage	Type	
E15	BBRX0_I_M	–	AI, AO	Baseband receiver input, channel 0, in-phase – minus
D15	BBRX0_I_P	–	AI, AO	Baseband receiver input, channel 0, in-phase – plus
C16	BBRX0_Q_M	–	AI, AO	Baseband receiver input, channel 0, quadrature-phase – minus
B16	BBRX0_Q_P	–	AI, AO	Baseband receiver input, channel 0, quadrature-phase – plus
B17	BBRX1_I_M	–	AI, AO	Baseband receiver input, channel 1, in-phase – minus
A17	BBRX1_I_P	–	AI, AO	Baseband receiver input, channel 1, in-phase – plus
E17	BBRX1_Q_M	–	AI, AO	Baseband receiver input, channel 1, quadrature-phase – minus
D17	BBRX1_Q_P	–	AI, AO	Baseband receiver input, channel 1, quadrature-phase – plus
T21	CSI0_A0_CLK_M	CSI	AI	MIPI CSI 0 (DPHY), differential clock – minus MIPI CSI 0 (CPHY), trio lane 0 – A
V21	CSI0_A1_LN1_P	CSI	AI	MIPI CSI 0 (DPHY), differential lane 1 – plus MIPI CSI 0 (CPHY), trio lane 1 – A
W21	CSI0_A2_LN2_M	CSI	AI	MIPI CSI 0 (DPHY), differential lane 2 – minus MIPI CSI 0 (CPHY), trio lane 2 – A
U20	CSI0_B0_LN0_P	CSI	AI	MIPI CSI 0 (DPHY), differential lane 0 – plus MIPI CSI 0 (CPHY), trio lane 0 – B
V22	CSI0_B1_LN1_M	CSI	AI	MIPI CSI 0 (DPHY), differential lane 1 – minus MIPI CSI 0 (CPHY), trio lane 1 – B
Y22	CSI0_B2_LN3_P	CSI	AI	MIPI CSI 0 (DPHY), differential lane 3 – plus MIPI CSI 0 (CPHY), trio lane 2 – B
U21	CSI0_C0_LN0_M	CSI	AI	MIPI CSI 0 (DPHY), differential lane 0 – minus MIPI CSI 0 (CPHY), trio lane 0 – C
W20	CSI0_C1_LN2_P	CSI	AI	MIPI CSI 0 (DPHY), differential lane 2 – plus MIPI CSI 0 (CPHY), trio lane 1 – C
Y21	CSI0_C2_LN3_M	CSI	AI	MIPI CSI 0 (DPHY), differential lane 3 – minus MIPI CSI 0 (CPHY), trio lane 2 – C
T22	CSI0_NC_CLK_P	CSI	AI	MIPI CSI 0 (D-PHY), differential clock – positive MIPI CSI 0 (CPHY), no connect
P6	CXO	PX11	DI	19.2 MHz clock input (sourced from PMIC)
AG4	DS_EN	PX_0	DI, PD	PMIC sleep status from PMIC to SoC Active-high input 1: PMIC is in deep sleep mode 0: PMIC in active mode
V2	DSI0_CLK_M	DSI	AO	MIPI DSI 0 (DPHY), differential lane clock – minus
V1	DSI0_CLK_P	DSI	AO	MIPI DSI 0 (DPHY), differential lane clock – plus

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Table 2-2 SW6100-1/SW6100P-1 bottom pin descriptions – general pins (cont.)

Pin no.	Pin name	Pin characteristics		Functional description
		Voltage	Type	
V4	DSI0_LN0_M	DSI	AO	MIPI DSI 0 (DPHY), differential lane 0 – minus
V3	DSI0_LN0_P	DSI	AO	MIPI DSI 0 (DPHY), differential lane 0 – plus
W2	DSI0_LN1_M	DSI	AO	MIPI DSI 0 (DPHY), differential lane 1 – minus
W3	DSI0_LN1_P	DSI	AO	MIPI DSI 0 (DPHY), differential lane 1 – plus
U2	DSI0_LN2_M	DSI	AO	MIPI DSI 0 (DPHY), differential lane 2 – minus
U3	DSI0_LN2_P	DSI	AO	MIPI DSI 0 (DPHY), differential lane 2 – plus
T2	DSI0_LN3_M	DSI	AO	MIPI DSI 0 (DPHY), differential lane 3 – minus
T1	DSI0_LN3_P	DSI	AO	MIPI DSI 0 (DPHY), differential lane 3 – plus
AJ13	EBI0_CAL	VDDQ_LPDDR5	AI	LPDDR calibration pad for external resistor connection
E13	ETDAC_M	-	AI, AO	Envelope tracker DAC pin – minus
E12	ETDAC_P	–	AI, AO	Envelope tracker DAC pin – plus
R3	EUSB0_EDM	–	AI, AO	Embedded USB2 port – minus
R4	EUSB0_EDP	–	AI, AO	Embedded USB2 port – plus
C19	FBRX_I	–	AI, AO	Tx transmit feedback receiver baseband in-phase output
D19	FBRX_Q	–	AI, AO	Tx transmit feedback receiver baseband quadrature output
E10	GNSS_L1_I	–	AI	GNSS receiver baseband input, in-phase for L1 band
D10	GNSS_L1_Q	–	AI	GNSS receiver baseband input, quadrature for L1 band
B10	GNSS_L5_I	–	AI	GNSS receiver baseband input, in-phase for L5 band
C10	GNSS_L5_Q	–	AI	GNSS receiver baseband input, quadrature for L5 band
P23	JTAG_SRST_N	PX14	DI, PU	JTAG reset for debug
N20	JTAG_TCK	PX14	DI, PU	JTAG clock input
N21	JTAG_TDI	PX14	DI, PU: nppdkp	JTAG data input
P22	JTAG_TDO	PX14	DO–Z	JTAG data output
P21	JTAG_TMS	PX14	DI, PU: nppdkp	JTAG mode select input
P20	JTAG_TRST_N	PX14	DI, PU: nppdkp	JTAG reset
F8	MODE_0	PX14	DI, PD: nppukp	Mode control bit 0
E8	MODE_1	PX14	DI, PD: nppukp	Mode control bit 1
M1	PCIE0_REFCLK_M	–	AI, AO	PCIe0 Gen 3 reference clock – negative
L2	PCIE0_REFCLK_P	–	AI, AO	PCIe0 Gen 3 reference clock – positive
M2	PCIE0_RX0_M	–	AI, AO	PCIe0 Gen 3 receive – negative
L1	PCIE0_RX0_P	–	AI, AO	PCIe0 Gen 3 receive – positive
P1	PCIE0_TX0_M	–	AI, AO	PCIe0 Gen 3 transmit – negative
P2	PCIE0_TX0_P	–	AI, AO	PCIe0 Gen 3 transmit – positive

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Table 2-2 SW6100-1/SW6100P-1 bottom pin descriptions – general pins (cont.)

Pin no.	Pin name	Pin characteristics		Functional description
		Voltage	Type	
G9	PS_HOLD	PX14	DO	SoC boot status and power supply hold output from SoC to PMIC Active-high pin with internal weak PD 1: Successful SoC boot, PMIC to hold the power supplies 0: PMIC to reset or shutdown the power supplies
J1	REFGEN_REXT0	–	AI, AO	External calibration resistor for the reference PHY 0
R22	REFGEN_REXT1	–	AI, AO	External calibration resistor for the reference PHY 1
AH4	RESIN_N	PX0	DI	Hardware reset input to SoC (driven by PMIC) Active: Low input 0: SoC in reset state 1: SoC in active state
B8	RESOUT_N	PX3	DO	Reset output pin from SoC to external circuitry Active-Low output pin, follows RESIN_N pin This signal is asserted or de-asserted a few clock cycles after a change on RESIN_N or after a watchdog expiration 0: SoC asserting a reset to peripheral devices 1: SoC deasserting reset to peripheral devices
D7	SLEEP_CLK	PX14	DI	32.768 kHz sleep clock input (sourced from PMIC)
N18	SPMI0_CLK	PX0	B, PD: nppukp	SPMI interface CLK
P17	SPMI0_DATA	PX0	B, PD: nppukp	SPMI interface DATA
C14	TXDAC_I_M	–	AI, AO	TXDAC in-phase – minus
B14	TXDAC_I_P	–	AI, AO	TXDAC in-phase – plus
B12	TXDAC_Q_M	–	AI, AO	TXDAC quadrature – minus
C12	TXDAC_Q_P	–	AI, AO	TXDAC quadrature – plus
D14	TXDAC_REF	–	AI	Reference voltage for TXDAC circuits
AH23	ZQ0	VDDQ_LPDDR5	AI	Pin for ZQ0 calibration

NOTE GPIO pins can support multiple functions. To assign GPIOs to particular functions (such as the options listed in the preceding table), designers must identify all their application's requirements and map each GPIO to its function carefully avoiding conflicts in GPIO assignments. See Table 2-3 for a list of all supported functions for each GPIO.

NOTE Device designers must examine each GPIO's external connection and programmed configuration, and take steps necessary to avoid excessive leakage current. Combinations of the following factors must be controlled properly:

- GPIO configuration
  - Input vs. output
  - Pull-up or pull-down

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- External connections
  - Unused inputs
  - Connections to high-impedance (tri-state) outputs
  - Connections to external devices that may not be attached

To help designers define their products' GPIO assignments, QTI provides an Excel spreadsheet that includes a worksheet for all SW6100/SW6100P GPIOs (in numeric order), pin numbers, pin voltages, pull states, and available configurations.

Click the following link to download the *SW6100/SW6100P Pin Assignment/GPIO Configuration Spreadsheet (80-83950-1A)* from the website.

SW6100/SW6100P Pin Assignment and GPIO Configuration Spreadsheet

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Table 2-3 SW6100-1/SW6100P-1 bottom pin descriptions – general-purpose input/output ports

Pin no.	Pin name	Wakeup function	Pin characteristics		Functional description		
			Voltage	Type	General purpose	SSC	LPASS
AB22	GPIO_0	Yes	PX14	PD: nppukp	Configurable GPIO 0	–	–
AB21	GPIO_1	–	PX14	PD: nppukp	Configurable GPIO 1	–	–
AC22	GPIO_2	–	PX14	PD: nppukp	Configurable GPIO 2	–	–
AD22	GPIO_3	Yes	PX14	PD: nppukp	Configurable GPIO 3	–	–
AC20	GPIO_4	Yes	PX14	PD: nppukp	Configurable GPIO 4	–	–
AD20	GPIO_5	–	PX14	PD: nppukp	Configurable GPIO 5	–	–
AC21	GPIO_6	–	PX14	PD: nppukp	Configurable GPIO 6	–	–
AE22	GPIO_7	Yes	PX14	PD: nppukp	Configurable GPIO 7	–	–
AD21	GPIO_8	Yes	PX14	PD: nppukp	Configurable GPIO 8	–	–
AF20	GPIO_9	–	PX14	PD: nppukp	Configurable GPIO 9	–	–
AE21	GPIO_10	–	PX14	PD: nppukp	Configurable GPIO 10	–	–
AE20	GPIO_11	Yes	PX14	PD: nppukp	Configurable GPIO 11	–	–
AH21	GPIO_12	–	PX712	PD: nppukp	Configurable GPIO 12	–	–
AH22	GPIO_13	Yes	PX712	PD: nppukp	Configurable GPIO 13	–	–
A4	GPIO_14	Yes	PX703	PD: nppukp	Configurable GPIO 14	–	–
A3	GPIO_15	Yes	PX703	PD: nppukp	Configurable GPIO 15	–	–
H23	GPIO_16	Yes	PX14	PD: nppukp	Configurable GPIO 16	–	–
J21	GPIO_17	–	PX14	PD: nppukp	Configurable GPIO 17	–	–
J22	GPIO_18	Yes	PX14	PD: nppukp	Configurable GPIO 18	–	–
J20	GPIO_19	Yes	PX14	PD: nppukp	Configurable GPIO 19	–	–
AA20	GPIO_20	Yes	PX14	PD: nppukp	Configurable GPIO 20	–	–
AA21	GPIO_21	–	PX14	PD: nppukp	Configurable GPIO 21	–	–
AA22	GPIO_22	–	PX14	PD: nppukp	Configurable GPIO 22	–	–
AB20	GPIO_23	Yes	PX14	PD: nppukp	Configurable GPIO 23	–	–
AF22	GPIO_24	Yes	PX712	PD: nppukp	Configurable GPIO 24	–	–
AF21	GPIO_25	Yes	PX712	PD: nppukp	Configurable GPIO 25	–	–
AG22	GPIO_26	–	PX712	PD: nppukp	Configurable GPIO 26	–	–

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Table 2-3 SW6100-1/SW6100P-1 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin no.	Pin name	Wakeup function	Pin characteristics		Functional description		
			Voltage	Type	General purpose	SSC	LPASS
AG20	GPIO_27	Yes	PX712	PD: nppukp	Configurable GPIO 27	–	–
G5	GPIO_28	Yes	PX14	PD: nppukp	Configurable GPIO 28	SSC_30	–
F6	GPIO_29	–	PX14	PD: nppukp	Configurable GPIO 29	SSC_31	–
AG21	GPIO_30	Yes	PX712	PD: nppukp	Configurable GPIO 30	–	–
AF19	GPIO_31	Yes	PX712	PD: nppukp	Configurable GPIO 31	–	–
K23	GPIO_32	Yes	PX14	PD: nppukp	Configurable GPIO 32	–	–
M20	GPIO_33	Yes	PX711	PD: nppukp	Configurable GPIO 33	–	–
L23	GPIO_34	Yes	PX14	PD: nppukp	Configurable GPIO 34	–	–
M22	GPIO_35	Yes	PX711	PD: nppukp	Configurable GPIO 35	–	–
N23	GPIO_36	Yes	PX14	PD: nppukp	Configurable GPIO 36	–	–
M23	GPIO_37	Yes	PX14	PD: nppukp	Configurable GPIO 37	–	–
M21	GPIO_38	Yes	PX711	PU: nppdkp	Configurable GPIO 38	–	–
N22	GPIO_39	Yes	PX711	PD: nppukp	Configurable GPIO 39	–	–
B4	GPIO_40	Yes	PX703	PD: nppukp	Configurable GPIO 40	SSC_25	–
E4	GPIO_41	Yes	PX14	PD: nppukp	Configurable GPIO 41	–	–
E5	GPIO_42	Yes	PX14	PD: nppukp	Configurable GPIO 42	–	–
D6	GPIO_43	–	PX14	PD: nppukp	Configurable GPIO 43	–	–
D5	GPIO_44	–	PX14	PD: nppukp	Configurable GPIO 44	–	–
G22	GPIO_45	Yes	PX14	PD: nppukp	Configurable GPIO 45	–	–
G23	GPIO_46	–	PX14	PD: nppukp	Configurable GPIO 46	–	–
H20	GPIO_47	Yes	PX14	PD: nppukp	Configurable GPIO 47	–	–
H21	GPIO_48	Yes	PX14	PD: nppukp	Configurable GPIO 48	–	–
B20	GPIO_49	Yes	PX707	PD: nppukp	Configurable GPIO 49	–	–
E20	GPIO_50	–	PX707	PD: nppukp	Configurable GPIO 50	–	–
F19	GPIO_51	Yes	PX707	PD: nppukp	Configurable GPIO 51	–	–
F20	GPIO_52	Yes	PX707	PD: nppukp	Configurable GPIO 52	–	–
G20	GPIO_53	Yes	PX707	PD: nppukp	Configurable GPIO 53	–	–

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Table 2-3 SW6100-1/SW6100P-1 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin no.	Pin name	Wakeup function	Pin characteristics		Functional description		
			Voltage	Type	General purpose	SSC	LPASS
B21	GPIO_54	–	PX14	PD: nppukp	Configurable GPIO 54	–	–
C21	GPIO_55	Yes	PX14	PD: nppukp	Configurable GPIO 55	–	–
B23	GPIO_56	Yes	PX14	PD: nppukp	Configurable GPIO 56	–	–
C23	GPIO_57	–	PX14	PD: nppukp	Configurable GPIO 57	–	–
C22	GPIO_58	Yes	PX14	PD: nppukp	Configurable GPIO 58	–	–
B22	GPIO_59	–	PX14	PD: nppukp	Configurable GPIO 59	–	–
E21	GPIO_60	Yes	PX14	PD: nppukp	Configurable GPIO 60	–	–
D21	GPIO_61	–	PX14	PD: nppukp	Configurable GPIO 61	–	–
E23	GPIO_62	Yes	PX705	PD: nppukp	Configurable GPIO 62	–	–
D23	GPIO_63	–	PX705	PD: nppukp	Configurable GPIO 63	–	–
F22	GPIO_64	Yes	PX706	PD: nppukp	Configurable GPIO 64	–	–
F23	GPIO_65	–	PX706	PD: nppukp	Configurable GPIO 65	–	–
F21	GPIO_66	Yes	PX706	PD: nppukp	Configurable GPIO 66	–	–
E22	GPIO_67	–	PX706	PD: nppukp	Configurable GPIO 67	–	–
K22	GPIO_68	–	PX709	PD: nppukp	Configurable GPIO 68	–	–
L22	GPIO_69	Yes	PX709	PD: nppukp	Configurable GPIO 69	–	–
L20	GPIO_70	–	PX709	PD: nppukp	Configurable GPIO 70	–	–
L21	GPIO_71	–	PX709	PD: nppukp	Configurable GPIO 71	–	–
A5	GPIO_72	–	PX703	PD: nppukp	Configurable GPIO 72	SSC_24	–
AH20	GPIO_73	–	PX14	PD: nppukp	Configurable GPIO 73	–	–
AG19	GPIO_74	Yes	PX14	PD: nppukp	Configurable GPIO 74	–	–
AH19	GPIO_75	Yes	PX14	PD: nppukp	Configurable GPIO 75	–	–
AE19	GPIO_76	Yes	PX14	PD: nppukp	Configurable GPIO 76	–	–
AF18	GPIO_77	–	PX14	PD: nppukp	Configurable GPIO 77	–	–
E7	GPIO_78	Yes	PX14	PD: nppukp	Configurable GPIO 78	SSC_43	–
K21	GPIO_79	Yes	PX14	PD: nppukp	Configurable GPIO 79	–	–
B7	GPIO_80	Yes	PX14	PD: nppukp	Configurable GPIO 80	–	–

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Table 2-3 SW6100-1/SW6100P-1 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin no.	Pin name	Wakeup function	Pin characteristics		Functional description		
			Voltage	Type	General purpose	SSC	LPASS
C7	GPIO_81	Yes	PX14	PU: nppdkp	Configurable GPIO 81	–	–
A7	GPIO_82	Yes	PX14	PD: nppukp	Configurable GPIO 82	–	–
C8	GPIO_83	–	PX14	PD: nppukp	Configurable GPIO 83	–	–
B6	GPIO_84	Yes	PX14	PD: nppukp	Configurable GPIO 84	–	–
C6	GPIO_85	–	PX14	PD: nppukp	Configurable GPIO 85	–	–
A6	GPIO_86	Yes	PX14	PD: nppukp	Configurable GPIO 86	–	–
AE18	GPIO_87	–	PX14	PD: nppukp	Configurable GPIO 87	–	–
AH18	GPIO_88	–	PX14	PD: nppukp	Configurable GPIO 88	–	–
AG18	GPIO_89	Yes	PX14	PD: nppukp	Configurable GPIO 89	–	–
N3	GPIO_90	–	PX14	PD: nppukp	Configurable GPIO 90	–	LPASS_0
N4	GPIO_91	Yes	PX14	PD: nppukp	Configurable GPIO 91	–	LPASS_1
M4	GPIO_92	–	PX14	PD: nppukp	Configurable GPIO 92	–	LPASS_2
L5	GPIO_93	Yes	PX14	PD: nppukp	Configurable GPIO 93	–	LPASS_3
D4	GPIO_94	–	PX701	PD: nppukp	Configurable GPIO 94	–	LPASS_4
E3	GPIO_95	Yes	PX701	PD: nppukp	Configurable GPIO 95	–	LPASS_5
C4	GPIO_96	–	PX701	PD: nppukp	Configurable GPIO 96	–	LPASS_6
B5	GPIO_97	–	PX701	PD: nppukp	Configurable GPIO 97	–	LPASS_7
D3	GPIO_98	Yes	PX701	PD: nppukp	Configurable GPIO 98	–	LPASS_8
E2	GPIO_99	Yes	PX701	PD: nppukp	Configurable GPIO 99	–	LPASS_9
F4	GPIO_100	–	PX14	PD: nppukp	Configurable GPIO 100	–	LPASS_10
G4	GPIO_101	Yes	PX14	PD: nppukp	Configurable GPIO 101	–	LPASS_11
B3	GPIO_102	Yes	PX703	PD: nppukp	Configurable GPIO 102	SSC_42	–
H3	GPIO_103	Yes	PX702	PD: nppukp	Configurable GPIO 103	SSC_0	–
H2	GPIO_104	–	PX702	PD: nppukp	Configurable GPIO 104	SSC_1	–
G2	GPIO_105	Yes	PX702	PD: nppukp	Configurable GPIO 105	SSC_2	–
F1	GPIO_106	–	PX702	PD: nppukp	Configurable GPIO 106	SSC_3	–
J2	GPIO_107	Yes	PX14	PD: nppukp	Configurable GPIO 107	SSC_4	–

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Table 2-3 SW6100-1/SW6100P-1 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin no.	Pin name	Wakeup function	Pin characteristics		Functional description		
			Voltage	Type	General purpose	SSC	LPASS
L4	GPIO_108	–	PX14	PD: nppukp	Configurable GPIO 108	SSC_5	–
J3	GPIO_109	–	PX14	PD: nppukp	Configurable GPIO 109	SSC_6	–
K4	GPIO_110	Yes	PX14	PD: nppukp	Configurable GPIO 110	SSC_7	–
C1	GPIO_111	Yes	PX704	PD: nppukp	Configurable GPIO 111	SSC_8	–
D2	GPIO_112	–	PX704	PD: nppukp	Configurable GPIO 112	SSC_9	–
K3	GPIO_113	Yes	PX702	PD: nppukp	Configurable GPIO 113	SSC_10	–
G1	GPIO_114	Yes	PX702	PD: nppukp	Configurable GPIO 114	SSC_11	–
E1	GPIO_115	–	PX702	PD: nppukp	Configurable GPIO 115	SSC_12	–
D1	GPIO_116	Yes	PX702	PD: nppukp	Configurable GPIO 116	SSC_13	–
E6	GPIO_117	Yes	PX14	PD: nppukp	Configurable GPIO 117	SSC_14	–
F5	GPIO_118	Yes	PX14	PD: nppukp	Configurable GPIO 118	SSC_15	–
J4	GPIO_119	Yes	PX14	PD: nppukp	Configurable GPIO 119	SSC_16	–
K5	GPIO_120	Yes	PX14	PD: nppukp	Configurable GPIO 120	SSC_17	–
B1	GPIO_121	Yes	PX703	PD: nppukp	Configurable GPIO 121	SSC_18	–
C2	GPIO_122	Yes	PX703	PD: nppukp	Configurable GPIO 122	SSC_19	–
A2	GPIO_123	Yes	PX703	PD: nppukp	Configurable GPIO 123	SSC_20	–
B2	GPIO_124	–	PX703	PD: nppukp	Configurable GPIO 124	SSC_21	–
G3	GPIO_125	Yes	PX14	PD: nppukp	Configurable GPIO 125	SSC_22	–
F3	GPIO_126	–	PX14	PD: nppukp	Configurable GPIO 126	SSC_23	–
AF4	GPIO_127	Yes	PX714	PD: nppukp	Configurable GPIO 127	SSC_32	–
AH3	GPIO_128	–	PX714	PD: nppukp	Configurable GPIO 128	SSC_33	–
AG3	GPIO_129	–	PX714	PD: nppukp	Configurable GPIO 129	SSC_34	–
AE4	GPIO_130	Yes	PX714	PD: nppukp	Configurable GPIO 130	SSC_35	–
AH2	GPIO_131	Yes	PX714	PD: nppukp	Configurable GPIO 131	SSC_29	–
AG2	GPIO_132	–	PX714	PD: nppukp	Configurable GPIO 132	SSC_28	–
AF3	GPIO_133	–	PX714	PD: nppukp	Configurable GPIO 133	SSC_27	–
AF2	GPIO_134	Yes	PX715	PD: nppukp	Configurable GPIO 134	SSC_26	–

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Table 2-3 SW6100-1/SW6100P-1 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin no.	Pin name	Wakeup function	Pin characteristics		Functional description		
			Voltage	Type	General purpose	SSC	LPASS
AE3	GPIO_135	Yes	PX715	PD: nppukp	Configurable GPIO 135	SSC_37	–
AD3	GPIO_136	Yes	PX715	PD: nppukp	Configurable GPIO 136	–	–
AE2	GPIO_137	Yes	PX715	PD: nppukp	Configurable GPIO 137	–	–
AC3	GPIO_138	Yes	PX715	PD: nppukp	Configurable GPIO 138	–	–
AD2	GPIO_139	Yes	PX715	PD: nppukp	Configurable GPIO 139	SSC_38	–
AC2	GPIO_140	Yes	PX715	PD: nppukp	Configurable GPIO 140	SSC_39	–
AB2	GPIO_141	Yes	PX715	PD: nppukp	Configurable GPIO 141	SSC_40	–
AB3	GPIO_142	Yes	PX715	PD: nppukp	Configurable GPIO 142	SSC_41	–
AA3	GPIO_143	–	PX715	PD: nppukp	Configurable GPIO 143	–	–
AA2	GPIO_144	–	PX715	PD: nppukp	Configurable GPIO 144	–	–
Y4	GPIO_145	Yes	PX715	PD: nppukp	Configurable GPIO 145	SSC_36	–
Y2	GPIO_146	–	PX715	PD: nppukp	Configurable GPIO 146	–	–
G21	GPIO_147	–	PX707	PD: nppukp	Configurable GPIO 147	–	–
H22	GPIO_148	–	PX14	PD: nppukp	Configurable GPIO 148	–	–
F7	GPIO_149	–	PX14	PD: nppukp	Configurable GPIO 149	–	LPASS_12

NOTE For detailed information about GPIO multiplexing, see *SW6100/SW6100P Pin Assignment and GPIO Configuration Spreadsheet* (80-83950-1A).

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Table 2-4 SW6100-1/SW6100P-1 bottom pin descriptions – NC, ground, and power-supply pins

Pin no.	Pin name	Functional description
A22	VCC_EMMC	Power for eMMC – control
A19	VCCI_EMMC	Power for eMMC – core I/O
A21	VCCQ_EMMC	Power for memory I/O
G11	VDD_A_GNSS_ADC_0P9	Power for the GPS ADC circuits – 0.9 V
W18	VDD_A_CSI_0_0P9	Power for the camera serial interface (CSI) – 0.9 V circuits
W19	VDD_A_CSI_0_1P2	Power for the camera serial interface (CSI) – 1.2 V circuits
T4	VDD_A_DSI_0_0P9	Power for the display serial interface (DSI) – 0.9 V circuits
V6	VDD_A_DSI_0_1P2	Power for the display serial interface (DSI) – 1.2 V circuits
U5	VDD_A_DSI_0_PLL_0P9	Power for the PLL circuits of display interface
AC11, AC12, AC14, AC15, AF13, AG13	VDD_A_EBI_0	Power for EBI circuits (DDR memory controller of SoC)
AD12	VDD_A_EBI_0_PLL_0P9	Power for EBI PLL in DDR memory controller of SoC
T6	VDD_A_EUSB_0_0P9	Power for EUSB2 core circuitry
U7	VDD_A_EUSB_0_1P2	Power for EUSB2 1.2 V circuitry
R6	VDD_A_PCIE_0_0P9	Power for internal 0.9 V PCIe controller circuitry
U6	VDD_A_PCIE_0_1P2	Power for internal 1.2 V PCIe controller circuitry
P7	VDD_A_REFGEN_0P875_0	Reference voltage for the circuitry to transmit clocks across the SoC – 0.875 V
U19	VDD_A_REFGEN_0P875_1	Reference voltage for the circuitry to transmit clocks across the SoC – 0.875 V
R7	VDD_A_REFGEN_1P2_0	Reference voltage for the circuitry to transmit clocks across the SoC – 1.2 V
U17	VDD_A_REFGEN_1P2_1	Reference voltage for the circuitry to transmit clocks across the SoC – 1.2 V
F16	VDD_AL_BBRX	Power for analog circuits – low voltage for BBRX
F17	VDD_AL_FBRX	Power for analog circuits – low voltage for FBRX
AA5, AA8, AB5, AB8, AC5, AD5, AE5, AF5, AH5, W7, W8, Y8	VDD_APC0	Power for the application processor – Arm Cortex-A55
AC10, AC8, AD10, AD8, AE8, AF9, AH9	VDD_APC1	Power for the application processor – Arm Cortex-A78
AA10, AA13, J13, K13, L13, M13, M16, N13, N16, P13, P16, R13, T13, U13, V10, V13, W10, W13, Y10, Y13	VDD_CX	Power for SoC core logic domain
G7, H10, H7, J10, J5, J7, K10, K7, L10, L7, M10, M7, N10, P10, R10, T10, U10	VDD_LPI_CX	Sensor and LPASS island logic domain core power supply
AD13	VDD_D_EBI_0	Power for EBI circuits digital core (DDR memory controller of SoC)
G18, H16, H18, J16, J18, K16, K18, L16, L18	VDD_GFX	Power for graphical processing unit

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Table 2-4 SW6100-1/SW6100P-1 bottom pin descriptions – NC, ground, and power-supply pins (cont.)

Pin no.	Pin name	Functional description
AE11, AE12, AE13, AE14, AF12, AG12	VDD_IO_EBI_0	Power for the EBI I/O circuits
M5, M19, AB19, W4	VDD_IX	Power for intermediate circuits
AA12, AB12, H12, J12, K12, L12, M12, N12, P12, R12, T12, U12, V12, W12, Y12	VDD_MX_A	Power rail for always-ON memory
AA15, AA7, AB15, AB7, AC7, AD7, AE7, AF16, AF6, AF8, AH16, AH6, AH8, J15, K15, L15, M15, N15, P15, R15, T15, U15, W15, Y15, Y7	VDD_MX_C	Collapsible memory power rail for GPU/Camera/ Video/APC/Modem
H8, J8, K8, L8, M8, N8, P8, R8, T8, U8, V8, T17	VDD_LPI_MX	Power for sensor and LPASS island memory domain
G12	VDD_MX_A_TXDAC_ETDAC	Power for circuits related to envelope tracker DAC and TXDAC of RF – memory domain
AA17, AB14, AB17, AC17, AC19, AD17, AD19, AE17, AF17, AH17, V17, W17, Y17, Y18	VDD_NSP	Power for the internal neural signal processor
AC4, R17	VDD_PX0	Special pads for the RESET IN, deep sleep and SPMI
AD4, AF14	VDD_PX1	Power for the pad group 1 – DDR controller RESET circuitry
H9	VDD_PX3	Power for the pad group 3 – SDIO RESETs
P5	VDD_PX11	Power for the pad group 11 – CXO pads
H5, L19, Y19, AA4, M6	VDD_PX14	Power for the pad group 14 – 1.2 V GPIOs, JTAG, PS_HOLD, SLEEP_CLK
G8	VDD_PX701	Power for the pad group 701 – 1.2 V/1.8 V dual pads
N5	VDD_PX702	Power for the pad group 702 – 1.2 V/1.8 V dual pads
H4	VDD_PX703	Power for the pad group 703 – 1.2 V/1.8 V dual pads
K6	VDD_PX704	Power for the pad group 704 – 1.2 V/1.8 V dual pads
J19	VDD_PX705	Power for the pad group 705 – 1.2 V/1.8 V dual pads
K19	VDD_PX706	Power for the pad group 706 – 1.2 V/1.8 V dual pads
R19	VDD_PX707	Power for the pad group 707 – 1.2 V/1.8 V dual pads
N19	VDD_PX709	Power for the pad group 709 – 1.2 V/1.8 V dual pads
T19	VDD_PX711	Power for the pad group 711 – 1.2 V/1.8 V dual pads
AA19	VDD_PX712	Power for the pad group 712 – 1.2 V/1.8 V dual pads
AB4	VDD_PX714	Power for the pad group 714 – 1.2 V/1.8 V dual pads
W5	VDD_PX715	Power for the pad group 715 – 1.2 V/1.8 V dual pads

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Table 2-4 SW6100-1/SW6100P-1 bottom pin descriptions – NC, ground, and power-supply pins (cont.)

Pin no.	Pin name	Functional description
M18	VDD_QFPROM	Power for programming the internal QFPROM
H13	VDD_AH_TXDAC_ETDAC	Power for circuits related to envelope tracker DAC and TXDAC of RF – core domain
AJ21, AJ4, W23, Y1	LPDDR5_VDD1	Power for DDR memory core – 1.8 V
AJ10, AJ14, AJ22, AJ3, N1, U23	LPDDR5_VDD2H	Power for VDD2H (high voltage) rail for LPDDR5X memory controller
AJ12, R1, T23	LPDDR5_VDD2L	Power for VDD2H (Low voltage) rail for LPDDR5X memory controller
AA1, AA23, AJ18, AJ6	LPDDR5_VDDQ	Power for memory I/O
A8, A9, AJ11, H1, N6, R20	DNC	Do not connect
A1, A10, A11, A12, A13, A14, A15, A16, A18, A20, A23, AA6, AA9, AA11, AA14, AA16, AA18, AB1, AB6, AB11, AB13, AB16, AB18, AB23, AC1, AC6, AC9, AC13, AC16, AC18, AC23, AD1, AD6, AD9, AD11, AD14, AD15, AD16, AD18, AD23, AE1, AE6, AE9, AE10, AE15, AE16, AE23, AF1, AF7, AF10, AF11, AF15, AF23, AG1, AG5, AG6, AG7, AG8, AG9, AG10, AG11, AG14, AG15, AG16, AG17, AG23, AH1, AH7, AH10, AH11, AH12, AH13, AH14, AH15, AJ1, AJ2, AJ5, AJ7, AJ8, AJ9, AJ15, AJ16, AJ17, AJ19, AJ20, AJ23, B9, B11, B13, B15, B18, B19, C5, C9, C11, C13, C15, C17, C18, C20, D8, D9, D11, D12, D13, D16, D18, D20, D22, E9, E11, E14, E16, E18, E19, F2, F9, F10, F11, F12, F13, F14, F15, F18, G6, G19, H6, H11, H14, H17, H19, J6, J9, J11, J14, J17, J23, K1, K2, K9, K11, K14, K17, K20, L3, L6, L9, L11, L14, L17, M3, M9, M11, M14, M17, N2, N7, N9, N11, N14, N17, P3, P4, P9, P11, P14, P19, R2, R11, R14, R16, R18, R21, R23, T3, T7, T9, T11, T14, T16, T20, U1, U4, U9, U11, U14, U16, U22, V5, V7, V9, V11, V14, V16, V20, V23, W1, W6, W9, W11, W14, W16, W22, Y3, Y9, Y11, Y14, Y16, Y20, Y23	GND	SoC GND bottom

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2.3 Pin assignments – top

2.3.1 Pin map – top

A high-level view of the top pin assignments is shown in the following figure.

The text within the following figure is difficult to read when viewing an 8½ inch × 11 inch hard copy. Other viewing options are available and defined in Pin map – bottom.

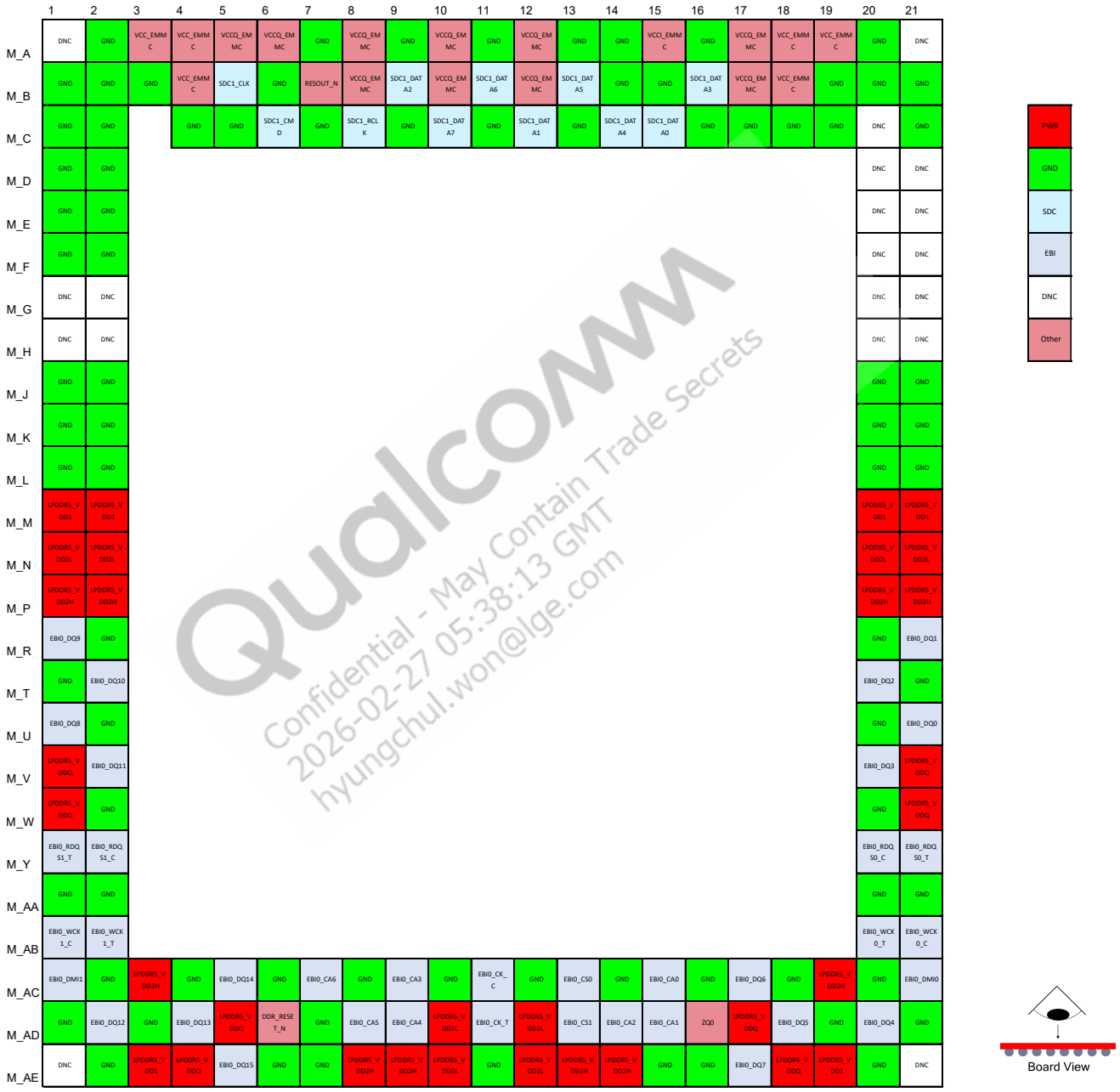

Figure 2-2 SW6100-1/SW6100P-1 top pin assignments

2.3.2 Pin descriptions – top

The top pins are described in Table 2-5 and Table 2-6.

Table 2-5 SW6100-1/SW6100P-1 top pin descriptions – general pins

Pin no.	Pin name	Pin characteristics		Functional description
		Voltage	Type	
M_AD6	DDR_RESET_N	PX1	DO, PU	LPDDR5 reset
M_AC15	EBI0_CA0	EBI	DO	EBI0 LPDDR5 command/address bit 0
M_AD15	EBI0_CA1	EBI	DO	EBI0 LPDDR5 command/address bit 1
M_AD14	EBI0_CA2	EBI	DO	EBI0 LPDDR5 command/address bit 2
M_AC9	EBI0_CA3	EBI	DO	EBI0 LPDDR5 command/address bit 3
M_AD9	EBI0_CA4	EBI	DO	EBI0 LPDDR5 command/address bit 4
M_AD8	EBI0_CA5	EBI	DO	EBI0 LPDDR5 command/address bit 5
M_AC7	EBI0_CA6	EBI	DO	EBI0 LPDDR5 command/address bit 6
M_AC11	EBI0_CK_C	EBI	DO	EBI0 LPDDR5 differential clock (C)
M_AD11	EBI0_CK_T	EBI	DO	EBI0 LPDDR5 differential clock (T)
M_AC13	EBI0_CS0	EBI	DO	EBI0 LPDDR5 chip – select 0
M_AD13	EBI0_CS1	EBI	DO	EBI0 LPDDR5 chip – select 1
M_AC21	EBI0_DMI0	EBI	DO	EBI0 LPDDR5 data mask for byte 0
M_AC1	EBI0_DMI1	EBI	DO	EBI0 LPDDR5 data mask for byte 1
M_U21	EBI0_DQ0	EBI	B	EBI0 LPDDR5 data bit 0
M_R21	EBI0_DQ1	EBI	B	EBI0 LPDDR5 data bit 1
M_T20	EBI0_DQ2	EBI	B	EBI0 LPDDR5 data bit 2
M_V20	EBI0_DQ3	EBI	B	EBI0 LPDDR5 data bit 3
M_AD20	EBI0_DQ4	EBI	B	EBI0 LPDDR5 data bit 4
M_AD18	EBI0_DQ5	EBI	B	EBI0 LPDDR5 data bit 5
M_AC17	EBI0_DQ6	EBI	B	EBI0 LPDDR5 data bit 6
M_AE17	EBI0_DQ7	EBI	B	EBI0 LPDDR5 data bit 7
M_U1	EBI0_DQ8	EBI	B	EBI0 LPDDR5 data bit 8
M_R1	EBI0_DQ9	EBI	B	EBI0 LPDDR5 data bit 9
M_T2	EBI0_DQ10	EBI	B	EBI0 LPDDR5 data bit 10
M_V2	EBI0_DQ11	EBI	B	EBI0 LPDDR5 data bit 11
M_AD2	EBI0_DQ12	EBI	B	EBI0 LPDDR5 data bit 12
M_AD4	EBI0_DQ13	EBI	B	EBI0 LPDDR5 data bit 13
M_AC5	EBI0_DQ14	EBI	B	EBI0 LPDDR5 data bit 14
M_AE5	EBI0_DQ15	EBI	B	EBI0 LPDDR5 data bit 15
M_Y20	EBI0_RDQS0_C	EBI	DI	EBI0 LPDDR5 differential read data strobe for byte 0 (C)
M_Y21	EBI0_RDQS0_T	EBI	B	EBI0 LPDDR5 differential read data strobe for byte 0 (T)
M_Y2	EBI0_RDQS1_C	EBI	DI	EBI0 LPDDR5 differential read data strobe for byte 1 (C)
M_Y1	EBI0_RDQS1_T	EBI	B	EBI0 LPDDR5 differential read data strobe for byte 1 (T)

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Table 2-5 SW6100-1/SW6100P-1 top pin descriptions – general pins (cont.)

Pin no.	Pin name	Pin characteristics		Functional description
		Voltage	Type	
M_AB21	EBI0_WCK0_C	EBI	DO	EBI0 LPDDR5 differential data clock for byte 0 (C)
M_AB20	EBI0_WCK0_T	EBI	DO	EBI0 LPDDR5 differential data clock for byte 0 (T)
M_AB1	EBI0_WCK1_C	EBI	DO	EBI0 LPDDR5 differential data clock for byte 1 (C)
M_AB2	EBI0_WCK1_T	EBI	DO	EBI0 LPDDR5 differential data clock for byte 1 (T)
M_B5	SDC1_CLK	PX3	DO, NP: pdpukp	eMMC secure digital controller clock
M_C6	SDC1_CMD	PX3	B, NP: pdpukp	eMMC secure digital controller command
M_C15	SDC1_DATA0	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 0
M_C12	SDC1_DATA1	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 1
M_B9	SDC1_DATA2	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 2
M_B16	SDC1_DATA3	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 3
M_C14	SDC1_DATA4	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 4
M_B13	SDC1_DATA5	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 5
M_B11	SDC1_DATA6	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 6
M_C10	SDC1_DATA7	PX3	B, NP: pdpukp	eMMC secure digital controller data bit 7
M_C8	SDC1_RCLK	PX3	DI, NP: pdpukp	eMMC secure digital controller return clock
M_AD16	ZQ0	VDDQ_LPDDR5	AI	Pin for ZQ0 calibration
M_B7	RESOUT_N	PX3	DO	Reset output from SoC for eMMC

Table 2-6 SW6100-1/SW6100P-1 top pin descriptions – NC, ground, and power-supply pins

Pin no.	Pin name	Functional description
M_AE3, M_AE19, M_M1, M_M2, M_M20, M_M21	LPDDR5_VDD1	Power for PoP LPDDR5 memory core 1 – VDD1
M_AC3, M_AC19, M_AE8, M_AE9, M_AE13, M_AE14, M_P1, M_P2, M_P20, M_P21	LPDDR5_VDD2H	Power for PoP LPDDR5 memory core 2 – VDD2H (High) – 1.05 V typical
M_AD10, M_AD12, M_AE10, M_AE12, M_N1, M_N2, M_N20, M_N21	LPDDR5_VDD2L	Power for PoP LPDDR5 memory core 2 – VDD2L (Low) – 0.9 V typical
M_AD5, M_AD17, M_AE4, M_AE18, M_V1, M_V21, M_W1, M_W21	LPDDR5_VDDQ	Power for PoP LPDDR5 memory IPO power – VDDQ – 0.6 V typical
M_A15	VCCI_EMMC	Power for eMMC – core I/O
M_A5, M_A6, M_A8, M_A10, M_A12, M_A17, M_B8, M_B10, M_B12, M_B17	VCCQ_EMMC	Power for memory I/O
M_A3, M_A4, M_A18, M_A19, M_B4, M_B18	VCC_EMMC	Power for eMMC – control

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Table 2-6 SW6100-1/SW6100P-1 top pin descriptions – NC, ground, and power-supply pins (cont.)

Pin no.	Pin name	Functional description
M_A1, M_A21, M_AE1, M_AE21, M_C20, M_D20, M_D21, M_E20, M_E21, M_F20, M_F21, M_G1, M_G2, M_G20, M_G21, M_H1, M_H2, M_H20, M_H21	DNC	Do not connect
M_A2, M_A7, M_A9, M_A11,M_A13, M_A14, M_A16, M_A20,M_AA1, M_AA2, M_AA20, M_AA21,M_AC2, M_AC4, M_AC6, M_AC8,M_AC10, M_AC12, M_AC14,M_AC16, M_AC18, M_AC20, M_AD1, M_AD3, M_AD7, M_AD19,M_AD21, M_AE2, M_AE6, M_AE7,M_AE11, M_AE15, M_AE16, M_AE20, M_B1, M_B2, M_B3, M_B6, M_B14, M_B15, M_B19, M_B20,M_B21, M_C1, M_C2, M_C4, M_C5,M_C7, M_C9, M_C11, M_C13,M_C16, M_C17, M_C18, M_C19,M_C21, M_D1, M_D2, M_E1, M_E2,M_F1, M_F2, M_J1, M_J2, M_J20,M_J21, M_K1, M_K2, M_K20,M_K21, M_L1, M_L2, M_L20, M_L21,M_R2, M_R20, M_T1, M_T21, M_U2,M_U20, M_W2, M_W20	GND	SoC GND top

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3 Electrical specifications

3.1 Absolute maximum ratings

The absolute maximum ratings (Table 3-1) reflect the stress levels that, if exceeded, may cause permanent damage to the device. No functionality is guaranteed outside the operating specifications. Functionality and reliability are only guaranteed within the operating conditions described in Operating conditions.

Table 3-1 Absolute maximum ratings

Power supply	Description	Min	Max	Unit
VDD_A_GNSS_ADC_0P9	Power for the GPS ADC circuits – 0.9 V	-0.3	1.01	V
VDD_A_CSI_0_0P9	Power for the camera serial interface (CSI) – 0.9 V circuits	-0.3	1.01	V
VDD_A_CSI_0_1P2	Power for the camera serial interface (CSI) – 1.2 V circuits	-0.3	1.39	V
VDD_A_DSI_0_0P9	Power for the display serial interface (DSI) – 0.9 V circuits	-0.3	1.01	V
VDD_A_DSI_0_1P2	Power for the display serial interface (DSI) – 1.2 V circuits	-0.3	1.39	V
VDD_A_DSI_0_PLL_0P9	Power for the PLL circuits of display interface	-0.3	1.01	V
VDD_A_EBI_0	Power for EBI circuits (DDR memory controller of SoC)	-0.3	1.16	V
VDD_A_EBI_0_PLL_0P9	Power for EBI PLL in DDR memory controller of SoC	-0.3	1.16	V
VDD_A_EUSB_0_0P9	Power for EUSB2 core circuitry	-0.3	1.05	V
VDD_A_EUSB_0_1P2	Power for EUSB2 1.2 V circuitry	-0.3	1.45	V
VDD_A_PCIE_0_0P9	Power for internal 0.9 V PCIe controller circuitry	-0.3	1.02	V
VDD_A_PCIE_0_1P2	Power for internal 1.2 V PCIe controller circuitry	-0.3	1.43	V
VDD_A_REFGEN_0P875_0	Reference voltage for the circuitry to transmit clocks across the SoC – 0.875 V	-0.3	1.02	V
VDD_A_REFGEN_0P875_1	Reference voltage for the circuitry to transmit clocks across the SoC – 0.875 V	-0.3	1.02	V
VDD_A_REFGEN_1P2_0	Reference voltage for the circuitry to transmit clocks across the SoC – 1.2 V	-0.3	1.38	V
VDD_A_REFGEN_1P2_1	Reference voltage for the circuitry to transmit clocks across the SoC – 1.2 V	-0.3	1.38	V
VDD_AL_BBRX	Power for analog circuits: Low voltage for BBRX	-0.3	1.43	V
VDD_AL_FBRX	Power for analog circuits: Low voltage for FBRX	-0.3	1.43	V
VDD_APC0	Power for the application processor – Arm Cortex-A55	-0.3	1.38	V
VDD_APC1	Power for the application processor – Arm Cortex-A78	-0.3	1.38	V
VDD_CX	Power for SoC core logic domain	-0.3	1.27	V
VDD_LPI_CX	Sensor and LPASS island logic domain core power supply	-0.3	1.16	V
VDD_D_EBI_0	Power for EBI circuits digital core (DDR memory controller of SoC)	-0.3	0.99	V
VDD_GFX	Power for graphical processing unit	-0.3	1.27	V
VDD_IO_EBI_0	Power for the EBI I/O circuits	-0.3	0.63	V
VDD_IX	Power for intermediate circuits	-0.3	1.04	V
VDD_MX_A	Power rail for always-ON memory	-0.3	1.16	V
VDD_MX_C	Collapsible Memory power rail for GPU/Camera/Video/APC/Modem	-0.3	1.16	V

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Table 3-1 Absolute maximum ratings (cont.)

Power supply	Description	Min	Max	Unit
VDD_LPI_MX	Power for sensor and LPASS island memory domain	-0.3	1.16	V
VDD_MX_A_TXDAC_ETDAC	Power for circuits related to envelope tracker DAC and TXDAC of RF – memory domain	-0.3	1.16	V
VDD_NSP	Power for the internal neural signal processor	-0.3	1.27	V
VDD_PX0	Special pads for the RESET IN, deep sleep and SPMI	-0.3	1.45	V
VDD_PX1	Power for the pad group 1: DDR controller RESET circuitry	-0.3	1.29	V
VDD_PX3	Power for the pad group 3: SDIO RESETs	-0.3	2.15	V
VDD_PX11	Power for the pad group 11: CXO pads	-0.3	1.45	V
VDD_PX14	Power for the pad group 14: 1.2 V GPIOs, JTAG, PS_HOLD, SLEEP_CLK	-0.3	1.45	V
VDD_PX701	Power for the pad group 701: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX702	Power for the pad group 702: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX703	Power for the pad group 703: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX704	Power for the pad group 704: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX705	Power for the pad group 705: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX706	Power for the pad group 706: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX707	Power for the pad group 707: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX709	Power for the pad group 709: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX711	Power for the pad group 711: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX712	Power for the pad group 712: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX714	Power for the pad group 714: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_PX715	Power for the pad group 715: 1.2 V/1.8 V dual pads	-0.3	1.45	V
		-0.3	2.18	V
VDD_QFPROM	Power for programming the internal QFPROM	-0.3	2.15	V
VDD_AH_TXDAC_ETDAC	Power for circuits related to envelope tracker DAC and TXDAC of RF – core domain	-0.3	2.09	V
T_S	Storage temperature [a b]	-55	150	°C

[a] The storage temperature range applies when the device is in the unassembled state (the device is not assembled in any PCB and is not electrically connected to any voltage or I/O signal. Damage may occur when the device is subjected to this temperature for any length of time.

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[b] For devices shipped in tape and reel, the storage temperature range is [+15℃ to +35℃] and < than 90% relative humidity (RH). QTI recommends allowing device to return to ambient room temperature before storage.

3.2 Operating conditions

Operating conditions include design team-controlled parameters such as power supply voltage, power distribution impedances, and thermal conditions. The SW6100/SW6100P meets the performance specifications, when used within the operating conditions, unless otherwise noted in those sections (provided the absolute maximum ratings have never been exceeded).

Customers should keep the thermal mitigation algorithm enabled with **default limits** to ensure that the operating temperature range is kept within the specification.

Table 3-2 Operating conditions for voltage rails

Power supply	Description	Min	Typ	Max	Unit
VDD_A_GNSS_ADC_0P9	Power for the GPS ADC circuits – 0.9 V	0.83	0.875	0.92	V
VDD_A_CSI_0_0P9	Power for the camera serial interface (CSI) – 0.9 V circuits	0.83	0.875	0.92	V
VDD_A_CSI_0_1P2	Power for the camera serial interface (CSI) – 1.2 V circuits	1.14	1.2	1.26	V
VDD_A_DSI_0_0P9	Power for the display serial interface (DSI) – 0.9 V circuits	0.83	0.875	0.92	V
VDD_A_DSI_0_1P2	Power for the display serial interface (DSI) – 1.2 V circuits	1.14	1.2	1.26	V
VDD_A_DSI_0_PLL_0P9	Power for the PLL circuits of display interface	0.83	0.875	0.92	V
VDD_A_EUSB_0_0P9	Power for EUSB2 core circuitry	0.72	0.75	0.95	V
VDD_A_EUSB_0_1P2	Power for EUSB2 1.2 V circuitry	1.08	1.2	1.32	V
VDD_A_PCIE_0_0P9	Power for internal 0.9 V PCIe controller circuitry	0.83	0.875	0.92	V
VDD_A_PCIE_0_1P2	Power for internal 1.2 V PCIe controller circuitry	1.14	1.2	1.26	V
VDD_A_REFGEN_0P875_0	Reference voltage for the circuitry to transmit clocks across the SoC – 0.875 V	0.825	0.875	0.925	V
VDD_A_REFGEN_0P875_1	Reference voltage for the circuitry to transmit clocks across the SoC – 0.875 V	0.825	0.875	0.925	V
VDD_A_REFGEN_1P2_0	Reference voltage for the circuitry to transmit clocks across the SoC – 1.2 V	1.14	1.2	1.26	V
VDD_A_REFGEN_1P2_1	Reference voltage for the circuitry to transmit clocks across the SoC – 1.2 V	1.14	1.2	1.26	V
VDD_AL_BBRX	Power for analog circuits: Low voltage for BBRX	1.1	1.225	1.3	V
VDD_AL_FBRX	Power for analog circuits: Low voltage for FBRX	1.1	1.225	1.3	V
VDD_D_EBI_0	Power for EBI circuits digital core (DDR memory controller of SoC)	0.352	0.728	1.155	V
VDD_IO_EBI_0	Power for the EBI I/O circuits	0.27	0.5	0.57	V
VDD_IX	Power for intermediate circuits	0.855	0.9	0.945	V
VDD_MX_A_TXDAC_ETDAC	Power for circuits related to envelope tracker DAC and TXDAC of RF – memory domain	0.504	0.752	1.05	V
VDD_PX0	Special pads for the RESET IN, deep sleep and SPMI	1.1	1.2	1.3	V
VDD_PX1	Power for the pad group 1: DDR controller RESET circuitry	1.01	1.1	1.17	V

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Table 3-2 Operating conditions for voltage rails (cont.)

Power supply	Description	Min	Typ	Max	Unit
VDD_PX3	Power for the pad group 3: SDIO RESETs	1.65	1.8	1.95	V
VDD_PX11	Power for the pad group 11: CXO pads	1.1	1.2	1.3	V
VDD_PX14	Power for the pad group 14: 1.2 V GPIOs, JTAG, PS_HOLD, SLEEP_CLK	1.1	1.2	1.3	V
VDD_PX701	Power for the pad group 701: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX702	Power for the pad group 702: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX703	Power for the pad group 703: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX704	Power for the pad group 704: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX705	Power for the pad group 705: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX706	Power for the pad group 706: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX707	Power for the pad group 707: 1.2 V/1.8 V dual pads	1.2	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX709	Power for the pad group 709: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX711	Power for the pad group 711: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX712	Power for the pad group 712: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX714	Power for the pad group 714: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_PX715	Power for the pad group 715: 1.2 V/1.8 V dual pads	1.1	1.2	1.3	V
		1.65	1.8	1.95	V
VDD_QFPROM	Power for programming the internal QFPROM	1.65	1.8	1.95	V
VDD_AH_TXDAC_ETDAC	Power for circuits related to envelope tracker DAC and TXDAC of RF – core domain	1.7	1.8	1.9	V
Thermal condition					
T	Device operating conditions	Tambient = -30	–	Tjunction = 95	℃

Table 3-3 Operating conditions for voltage rails with AVS

VDD rail	Mode	Min	Max	Unit
VDD_CX	Turbo_L1	0.65	0.964	V
	Turbo	0.61	0.907	V
	Nominal	0.56	0.823	V

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Table 3-3 Operating conditions for voltage rails with AVS (cont.)

VDD rail	Mode	Min	Max	Unit
	SVS_L1	0.525	0.797	V
	SVS	0.495	0.784	V
	LowSVS	0.495	0.713	V
	LowSVS_D1	0.495	0.678	V
VDD_MXA / VDD_MXC	Turbo_L1	0.75	0.942	V
	Turbo	0.75	0.907	V
	Nominal_L1	0.705	0.863	V
	Nominal	0.705	0.854	V
	SVS_L1	0.705	0.797	V
	SVS	0.675	0.797	V
	LowSVS	0.675	0.757	V
	LowSVS_D1	0.675	0.757	V
	LowSVS_D2	0.675	0.757	V
	LowSVS_D3	0.675	0.757	V
VDD_GFX	Turbo_L1	0.65	0.964	V
	Turbo	0.61	0.902	V
	Nominal_L1	0.59	0.858	V
	Nominal	0.56	0.818	V
	SVS_L1	0.525	0.790	V
	SVS	0.49	0.772	V
	LowSVS	0.465	0.703	V
	LowSVS_D1	0.465	0.668	V
	LowSVS_D2	0.45	0.637	V
	LowSVS_D3	0.45	0.602	V
VDD_NSP	Turbo_L1	0.65	0.991	V
	Turbo	0.61	0.948	V
	Nominal_L1	0.59	0.897	V
	Nominal	0.56	0.843	V
	SVS_L1	0.525	0.79	V
	SVS	0.495	0.768	V
	LowSVS	0.495	0.707	V
	LowSVS_D1	0.495	0.672	V
	LowSVS_D2	0.495	0.638	V
VDD_LPI_CX	Turbo	0.61	0.902	V
	Nominal	0.56	0.823	V
	SVS_L1	0.525	0.7972	V
	SVS	0.495	0.7796	V
	LowSVS	0.495	0.718	V
	LowSVS_D1	0.495	0.6828	V

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Table 3-3 Operating conditions for voltage rails with AVS (cont.)

VDD rail	Mode	Min	Max	Unit
VDD_LPI_MX	Turbo	0.75	0.902	V
	Nominal	0.705	0.858	V
	SVS_L1	0.705	0.797	V
	SVS	0.705	0.797	V
	LowSVS	0.705	0.797	V
	LowSVS_D1	0.705	0.797	V
VDDA_EBI	Turbo_L1	0.77	0.920	V
	Turbo	0.77	0.920	V
	Nominal	0.67	0.841	V
	SVS	0.62	0.823	V
	LowSVS	0.55	0.744	V
	LowSVS_D1	0.55	0.744	V
VDD_APC0	Turbo_L1	0.75	0.894	V
	Turbo	0.654	0.842	V
	Nominal_L1	0.59	0.802	V
	Nominal	0.56	0.766	V
	SVS_L1	0.525	0.742	V
	SVS	0.495	0.73	V
	LowSVS	0.48	0.666	V
	LowSVS_D1	0.48	0.634	V
	LowSVS_D2	0.48	0.606	V
VDD_APC1	Turbo_L1	0.774	0.894	V
	Turbo	0.774	0.842	V
	Nominal_L1	0.61	0.802	V
	Nominal	0.582	0.766	V
	SVS_L1	0.546	0.742	V
	SVS	0.51	0.73	V
	LowSVS	0.51	0.666	V
	LowSVS_D1	0.51	0.634	V
	LowSVS_D2	0.51	0.606	V

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3.3 Power delivery network specification

A detailed power delivery network specification is available in the *SW6100/SW6100P Hardware Design Guide (80-83950-5)*.

3.4 DC power characteristics

3.4.1 Average operating current

Detailed current consumption information and details about the operating modes tested are available in the *SW6100 Linux Wear (LW1.0) Current Consumption Data Application Note (80-83950-7)* and *SW6100 Linux Android Wear (LAW1.0) Current Consumption Data Application Note (80-83950-7A)*.

3.4.2 Dhrystone and rock-bottom maximum power

Table 3-4 Dhrystone and rock bottom maximum power

SoC Designation	1 × Arm Cortex-A78 (Kryo Gold) + 4 × A55 (Kryo Silver) 95℃ (Tj) [a b c]	Rock bottom mW 30°C (Tj) [d]
SW6100	1.1 W at Rail	6 mW at Rail

[a] Dhrystone specification applies to SW6100 CS devices that run at 2100 MHz (1 × Arm Cortex-A78 Gold), and 1950 MHz (4 × Arm Cortex-A55 Silver).
[b] Dhrystone power should be measured on VDD_APC0 and VDD_APC1 rails, at the point right before PDN capacitors (with a small resistor in serial inserted, if necessary).
[c] Measurement sampling rate should be > 1.25 Msps (or < 0.8 µS), and average window should be > 1 ms (or > 1250 samples).
[d] Rock Bottom (VDD_LPI_CX, VDD_LPI_MX, VDD_MXA) should be measured at the VDD_LPI_CX, VDD_LPI_MX, VDD_MXA rails. See *SW6100 Linux Wear (LW1.0) Current Consumption Data Application Note* (80-83950-7) for the test setup.

3.5 Power sequencing

The PMIC includes power-on circuits that provide the proper power sequencing for the entire SW6100 chipset. The supplies are turned on as groups of regulators that are selected by the hardware configuration of certain PMIC pins. For more information, see the PMIC Data sheets, *PW6100-1 Power Management IC Data Sheets (80-83961-1)* and *PW6100-0 Power Management IC Data Sheet (80-83960-1)*.

A high-level summary of the required power-on sequence is shown below:

- VDD_PX0, VDD_EUSB_1P2
- VDD_MX_A
- VDD_MX_C
- VDD_LPI_MX
- VDD_A_EBI_0, VDD_A_EBI_0_PLL_0P9
- VDD_CX, VDD_D_EBI_0
- VDD_LPI_CX
- VDD_PX14, VDD_PX709_1P2
- VDD_IX
- VDD_PX7xx_1P8, VDD_PX3
- VDD_PX11, VDD_A_CSI_0_1P2, VDD_A_DSI_0_1P2, VDD_A_PCIE_0_1P2, VDD_A_EUSB_0_1P2, VDD_A_REFGEN_1P2_0/1
- LPDDR5_VDD1
- VDD_PX1, LPDDR5_VDD2H
- LPDDR5_VDD2L
- VDD_IO_EBI_0, LPDDR5_VDDQ
- VDD_A_CSI_0_0P9, VDD_A_DSI_0_0P9, VDD_A_DSI_0_PLL_0P9, VDD_A_PCIE_0_0P9, VDD_A_EUSB_0_0P9, VDD_A_REFGEN_0P875_0/1, VDD_A_GNSS_ADC_0P9

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- VDD_EUSB_3P0
- VCC_EMMC
- VCCQ_EMMC
- VDD_APC0

Comments regarding this sequence:

- Other noncritical supplies are included within the power-on sequence. Any other supplies can be powered on by software after the sequence is completed.
- Each domain must reach its 90% of its value before the next domain starts ramping up. For example, when VDD_CX reaches 90% of its value, the VDD_PX supply can start ramping up.

3.6 Digital-logic characteristics

A digital I/Os performance specification depends on its pad type, its usage, and/or its supply voltage:

- Some digital I/Os are dedicated for interconnections between the SW6100/SW6100P chipsets and other ICs within the QTI chipset; therefore, specifications are not required.
- Some digital I/Os are defined by existing standards, such as I²C and SPI. QTI devices comply with those standards; therefore, additional specifications are not required.
- All other digital I/Os require performance specifications.

Table 3-5 DC specification of eSPMI

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	200	–	mV
I_{IH}	Input high leakage current	–	–	µA
I_{IL}	Input low leakage current	–	–	µA
IOZH	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	0.8 × VDD_PX	0.2 × VDD_PX	V
V_{OL}	Low-level output voltage, CMOS	0.8 × VDD_PX	0.2 × VDD_PX	V
I_{OLH}	Output current drive	2	2	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

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Table 3-6 DC specification of 1.2 V GPIOs

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	300	–	mV
I_{IH}	Input high leakage current	–	5	µA
I_{IL}	Input low leakage current	-5	–	µA
I_{OZH}	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	0.75 × VDDPX	0.25 × VDDPX	V
V_{OL}	Low-level output voltage, CMOS	0.75 × VDDPX	0.25 × VDDPX	V
I_{OLH}	Output current drive (programmable HDRIVE: 000 to 111)	1.33	10.66	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

Table 3-7 DC specification of 1.8 V GPIOs

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	300	–	mV
I_{IH}	Input high leakage current	–	5	µA
I_{IL}	Input low leakage current	-5	–	µA
I_{OZH}	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	VDD_PX- 0.45	0.45	V
V_{OL}	Low-level output voltage, CMOS	VDD_PX- 0.45	0.45	V

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Table 3-7 DC specification of 1.8 V GPIOs (cont.)

Parameter	Description	Min	Max	Unit
I_{OLH}	Output current drive (programmable HDRIVE: 000 to 111)	2	16	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

Table 3-8 DC specification of I²C 1.2 V/1.8 V

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	200	–	mV
I_{IH}	Input high leakage current	–	10	µA
I_{IL}	Input low leakage current	-10	–	µA
I_{OZH}	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	–	0.2 × VDD_PX	V
V_{OL}	Low-level output voltage, CMOS	–	0.2 × VDD_PX	V
I_{OLH}	Output current drive	–	2	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

Table 3-9 DC specification of I³C 1.2 V

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	200	–	mV
I_{IH}	Input high leakage current	–	5	µA
I_{IL}	Input low leakage current	-5	–	µA
I_{OZH}	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ

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Table 3-9 DC specification of I³C 1.2 V (cont.)

Parameter	Description	Min	Max	Unit
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	VDD_PX - 0.18	0.18	V
V_{OL}	Low-level output voltage, CMOS	VDD_PX - 0.18	0.18	V
I_{OLH}	Output current drive	2	2	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

Table 3-10 DC specification of I³C 1.8 V

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-levelinput voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	200	–	mV
I_{IH}	Input high leakage current	–	10	µA
I_{IL}	Input low leakage current	-10	–	µA
I_{OZH}	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	VDD_PX - 0.27	0.27	V
V_{OL}	Low-level output voltage, CMOS	VDD_PX - 0.27	0.27	V
I_{OLH}	Output current drive	3	3	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

Table 3-11 DC specification of SLIMbus

Parameter	Description	Min	Max	Unit
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = low)	0.35 × VDD_PX	0.65 × VDD_PX	V
V_{IH}	High-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{IL}	Low-level input voltage, CMOS/Schmitt (HIHYS_EN = high)	0.3 × VDD_PX	0.7 × VDD_PX	V
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = low)	100	–	mV
V_{SHYS}	Schmitt hysteresis voltage (HIHYS_EN = high)	200	–	mV
I_{IH}	Input high leakage current	–	2	µA
I_{IL}	Input low leakage current	-2	–	µA
I_{OZH}	High-level, tri-state leakage current	–	–	µA
I_{OZL}	Low-level, tri-state leakage current	–	–	µA

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Table 3-11 DC specification of SLIMbus (cont.)

Parameter	Description	Min	Max	Unit
R_{PD}	Pull-down resistance	10	50	kΩ
R_{PU}	Pull-up resistance	10	50	kΩ
R_{KP}	Bus keeper resistance	10	50	kΩ
V_{OH}	High-level output voltage, CMOS	0.9 × VDD_PX	0.1 × VDD_PX	V
V_{OL}	Low-level output voltage, CMOS	0.9 × VDD_PX	0.1 × VDD_PX	V
I_{OLH}	Output current drive	1	1	mA

NOTE I_{IH}, I_{IL}, I_{OZH}, and I_{OZL} values are based on characterization across PVT.

3.7 Timing characteristics

Specifications for the device timing characteristics are included (where appropriate) under each function's section, along with all its other performance specifications. Some general comments about timing characteristics and pertinent pad design methodologies are included here.

NOTE All SW6100/SW6100P chipsets are characterized with actively terminated loads; therefore, all baseband timing parameters in this document assume no bus loading. This is described in more detail in Rise and fall time specifications.

3.7.1 Timing diagram conventions

The conventions used within timing diagrams throughout this document are shown in the following figure.

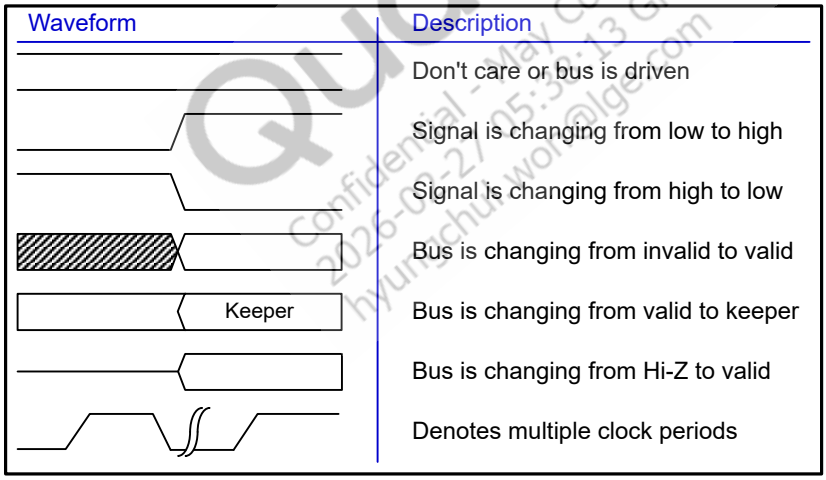

Figure 3-1 Timing diagram conventions

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For each signal in the diagram:

- One clock period (T) extends from one rising clock edge to the next rising clock edge.
- The high level represents 1, the low level represents 0, and the middle level represents the floating (high-impedance) state.
- When both the high and low levels are shown over the same time interval, the meaning depends on the signal type:
  - For a bus type signal (multiple bits), the processor or external interface is driving a value, but that value may or may not be valid.
  - For a single signal, this indicates don't care.

3.7.2 Rise and fall time specifications

The testers that characterize SW6100/SW6100P chipsets have actively terminated loads, making the rise and fall times quicker (mimicking a no-load condition). The impact that different external load conditions have on rise and fall times is shown in Figure 3-2.

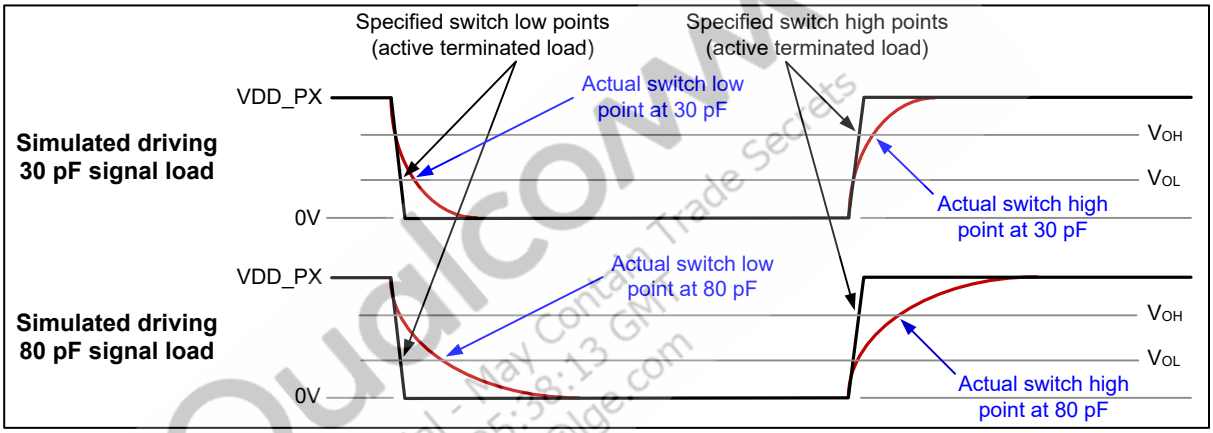

Figure 3-2 Rise and fall times under different load conditions

To account for external load conditions, rise or fall times must be added to parameters that start timing at the SW6100/SW6100P chipsets and terminate at an external device (or vice versa). Adding these rise and fall times is equivalent to applying capacitive load derating factors.

3.8 Memory support

The EBI0 port is dedicated to the PoP LPDDR5/5X SDRAM memory that is attached to the top of the SW6100/SW6100P chipsets.

3.9 Multimedia

Multimedia parameters requiring performance specification are addressed in this section.

3.9.1 Camera interfaces

The SW6100/SW6100P chipsets support up to 4-lane D-PHY and C-PHY camera interfaces.

Table 3-12 Supported MIPI_CSI standards and exceptions

Applicable standard	Feature exceptions
MIPI Alliance Specification for D-PHY v1.2	Supported up to 2.5 Gbps/lane
MIPI Alliance Specification for C-PHY v1.0	Supported up to 2.5 Gsps/Trio

3.9.2 Display support

The SW6100/SW6100P chipsets support up to one 4-lane D-PHY display.

Table 3-13 Supported MIPI_DSI standards and exceptions

Applicable standard	Feature exceptions
MIPI-DSI D-PHY v1.2 port	Supports up to 2.5 Gbps/lane

3.9.3 Audio support

The following audio interfaces are supported with SW6100/SW6100P chipsets:

- Digital microphone PDM interface
- SLIMbus interface
- PCM/TDM interfaces

3.10 Connectivity

The connectivity functions supported by the SW6100/SW6100P chipsets that require electrical specifications include:

- SD, multimedia cards (MMC)
- Digital microphone (PDM) interface
- SoundWire (SWR) interface
- Serial low-power inter-chip media bus (SLIMbus) interface
- Inter-IC sound (I^2S) interfaces
- Pulse-coded modulation (PCM) interfaces
- Time-division multiplexing (TDM) interfaces

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- Built-in USB physical layer (PHY)
- Through proper configuration of the QUP ports:
  - Universal asynchronous receiver/transmitter (UART) ports
  - Inter-integrated circuit interfaces (I²C and I³C)
  - Serial peripheral interface (SPI) ports
  - Dedicated I²C interfaces for camera (CCI I²C)

Pertinent specifications for these functions are detailed in the following subsections. In addition to the following hardware specifications, see the latest software release notes for software-based performance features or limitations.

3.10.1 SD interface

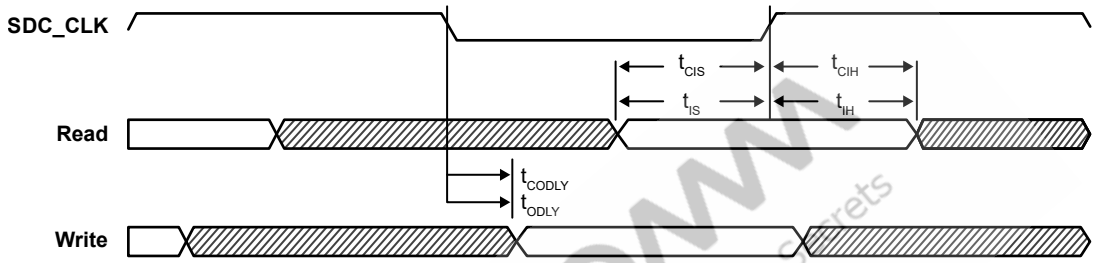

Figure 3-3 Single data rate – SDR mode

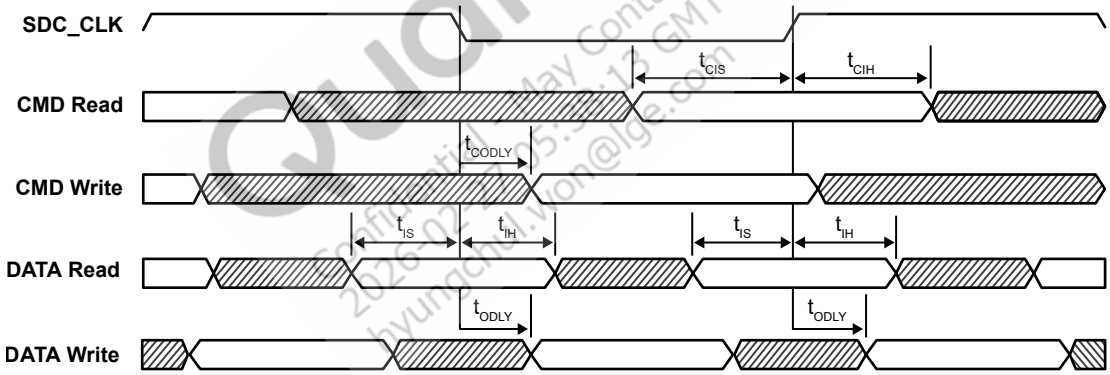

Figure 3-4 Double data rate – DDR mode

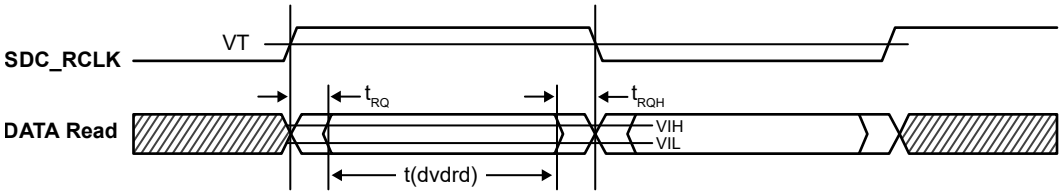

Figure 3-5 HS400 mode input timing

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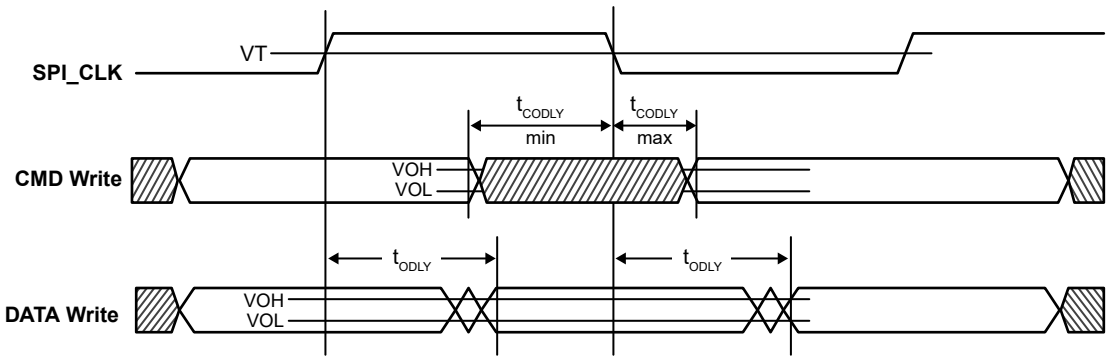

Figure 3-6 HS400 mode output timing

Table 3-14 SDC1 timing specification – Regular mode

SDCC modes	Frequency	Parameter	Description	Min (ns)	Max (ns)
HS400 (DDR200 MHz)	192 MHz	Tclhigh	Clock high time	2.292	2.917
		tCIDV	Command input valid window	2.4	–
		tCODLY	Command output delay	-1.45	0.85
		tRQ	DQS to DQ valid setup	0.4	–
		tRQH	DQS to DQ valid hold	0.4	–
		tIDV	Data input valid window	1.7	–
		tODLY	Data output delay	0.4	1.85
HS200 (SDR200 MHz)	192 MHz	Tclhigh	Clock high time	1.563	3.646
		tCIDV	Command input valid window	2.4	–
		tCODLY	Command output delay	-1.45	0.85
		tIDV	Data input valid window	2.4	–
		tODLY	Data output delay	-1.45	0.85

Table 3-15 SDC2 timing specification – Regular mode

SDCC modes	Frequency	Parameter	Description	Min (ns)	Max (ns)
SDR50	100 MHz	Tclhigh	Clock high time	3	7
		tCIDV	Command input valid window	2.4	–
		tCODLY	Command output delay	-3.7	1.5
		tIDV	Data input valid window	2.4	–
		tODLY	Data output delay	-3.7	1.5

NOTE SDC2 is for SDIO in the SW6100/SW6100P chipset, not SD 3.0.

3.10.2 USB interfaces

Table 3-16 Supported USB standards and exceptions

Applicable standard	Feature exceptions
Universal Serial Bus Specification, Revision 2.0 (August 11, 2014 or later)	Host mode and OTG is not supported

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3.10.3 UIM interface

Table 3-17 Supported UIM standards and exceptions

Applicable standard	Feature exceptions
ISO/IEC 7816-3	Class A sim cards are not supported

3.10.4 Digital microphone PDM interface

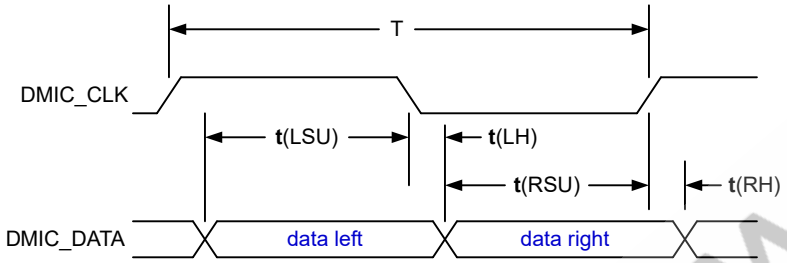

Figure 3-7 Digital microphone PDM interface timing

Table 3-18 Digital microphone timing

Parameter		Min	Typ	Max	Unit
1/T	DMIC clock frequency	0.6	–	12.288	MHz
	DMIC clock duty cycle	45	–	55	%
t(LSU)	Data left setup time to clock falling edge	10	–	–	ns
t(LH)	Data left hold time to clock falling edge	0	–	–	ns
t(RSU)	Data right setup time to clock rising edge	10	–	–	ns
t(RH)	Data right hold time to clock rising edge	0	–	–	ns

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3.10.5 SoundWire interface

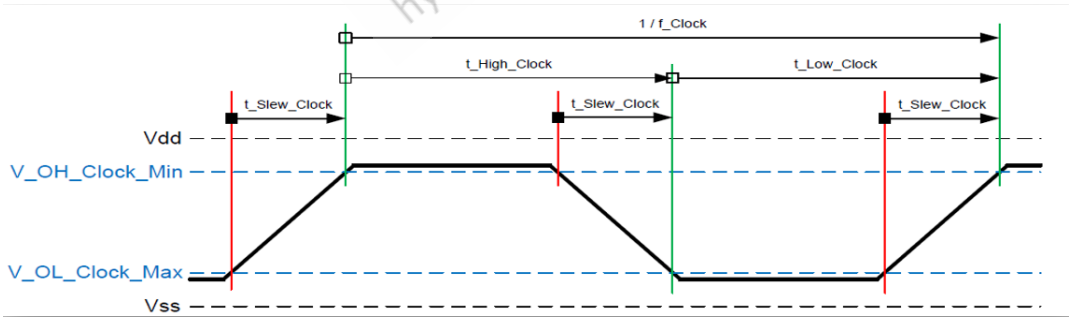

Figure 3-8 Duty cycle of clock output from the master

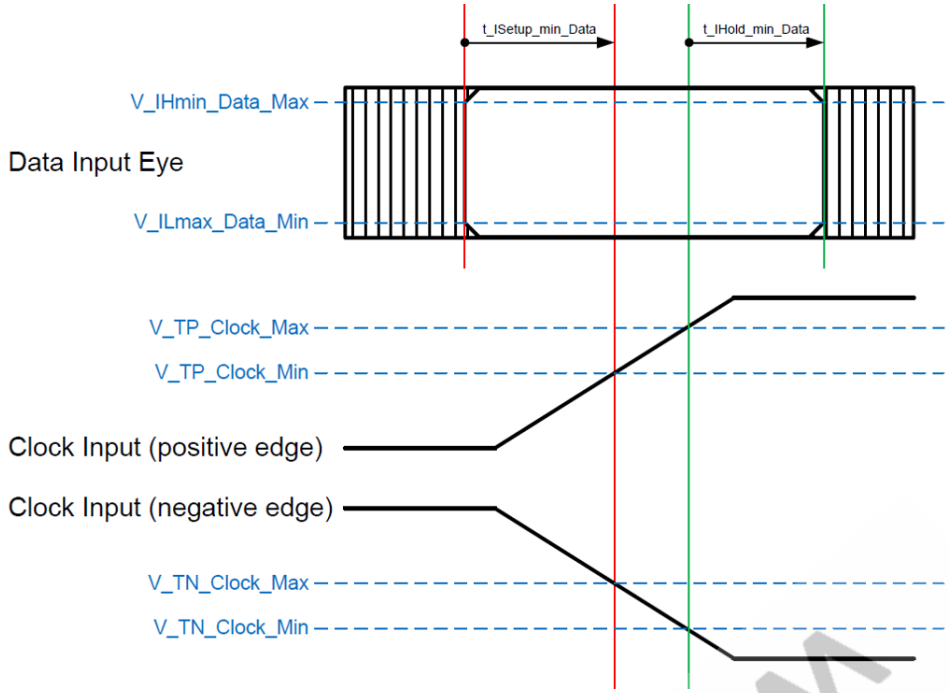

Figure 3-9 Setup and hold time

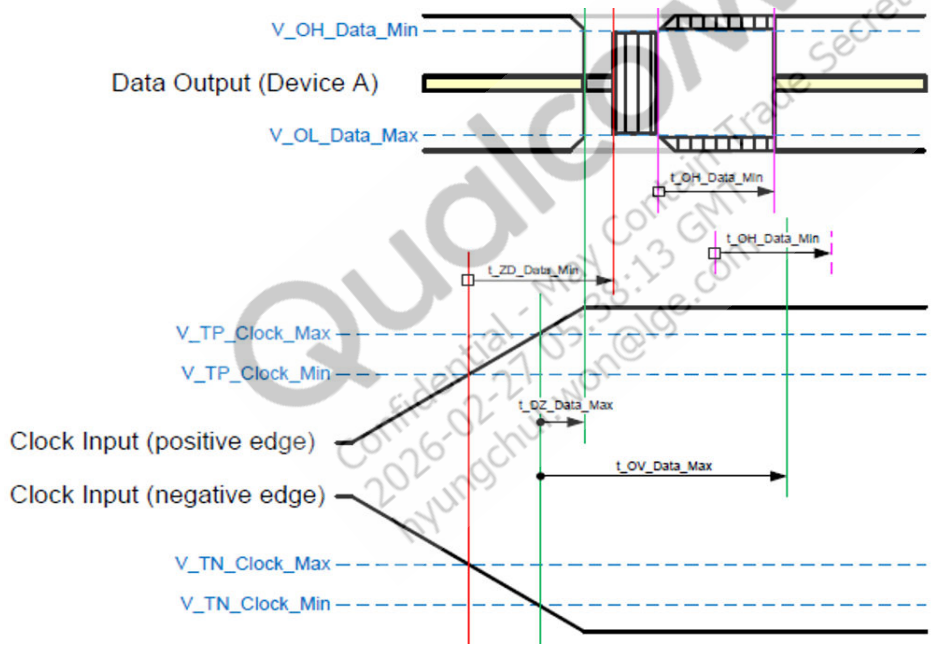

Figure 3-10 Output enable time (Toh), Hi-Z measurements (TZD and TDZ)

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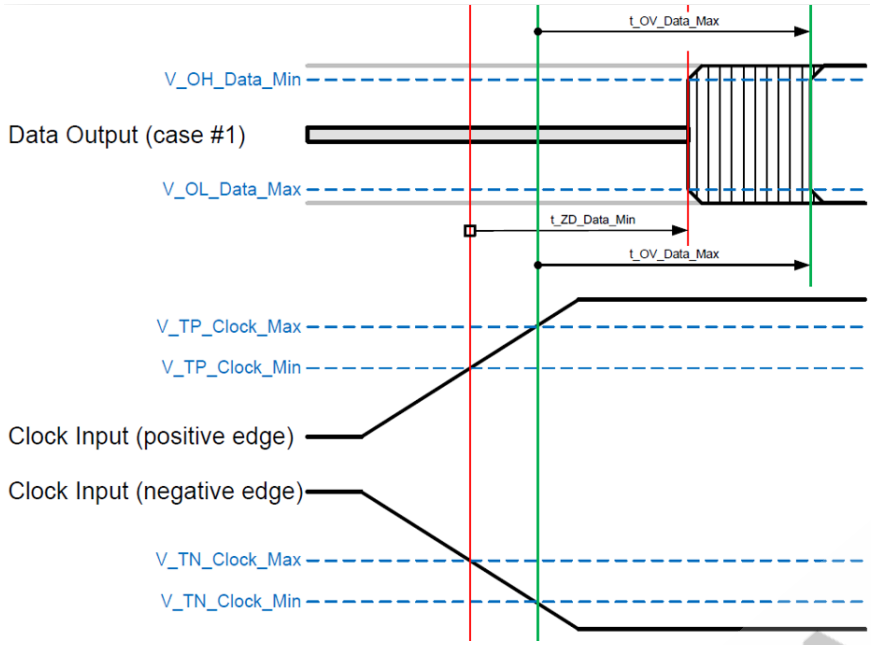

Figure 3-11 Output valid time (T_{ov})

Table 3-19 SoundWire timing specifications

Symbol	Parameter	Min	Max	Units
f_Clock_small	Frequency of clock signal in small systems	–	12.288	MHz
t_High_Clock_small	Duration of high half-period on clock output signal in small systems	35.3	–	ns
t_Low_Clock_small	Duration of low half-period on clock output signal in small systems	35.3	–	ns
t_DZ_Data	Time to disable data output signal after positive or negative edge on clock input signal	–	4	ns
t_ZD_Data	Time to enable data output signal after positive or negative edge on clock input signal	7.9	–	ns
t_OV_Data_small	Time to valid data output signal after positive or negative edge on clock input signal in small systems	–	27.6	ns
t_OH_Data	Time for data output signal to remain enabled and valid after first becoming valid	6.7	–	ns
t_ISetup_min_Data	Input setup time	0	–	ns
t_IHold_min_Data	Input hold time	4	–	ns
DC_Out_Clock	Duty cycle generated at clock output signal. Calculated from t_Low_Clock / (t_Low_Clock + t_High_Clock)	46% of the SWR CLK	54% of the SWR CLK	–

NOTE Characterization is performed at highest frequency 12.288 MHz only.

3.10.6 SLIMbus interface

Table 3-20 Supported SLIMbus standards and exceptions

Applicable standard	Feature exceptions
MIPI Alliance Specification for Serial Low-power Interchip Media Bus Version 2.0	None

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3.10.7 I²S interfaces timing diagam and specifications

There are two I²S interface types supported by the SW6100/SW6100P:

- Legacy I²S interfaces for primary and secondary microphones and speakers
- The multiple I²S (MI²S) interface for microphone and speaker functions The following information applies to both interface types.

Table 3-21 Supported I²S standards and exceptions

Applicable standards	Feature exceptions
Philips I2S Bus Specifications revised June 5, 1996	None

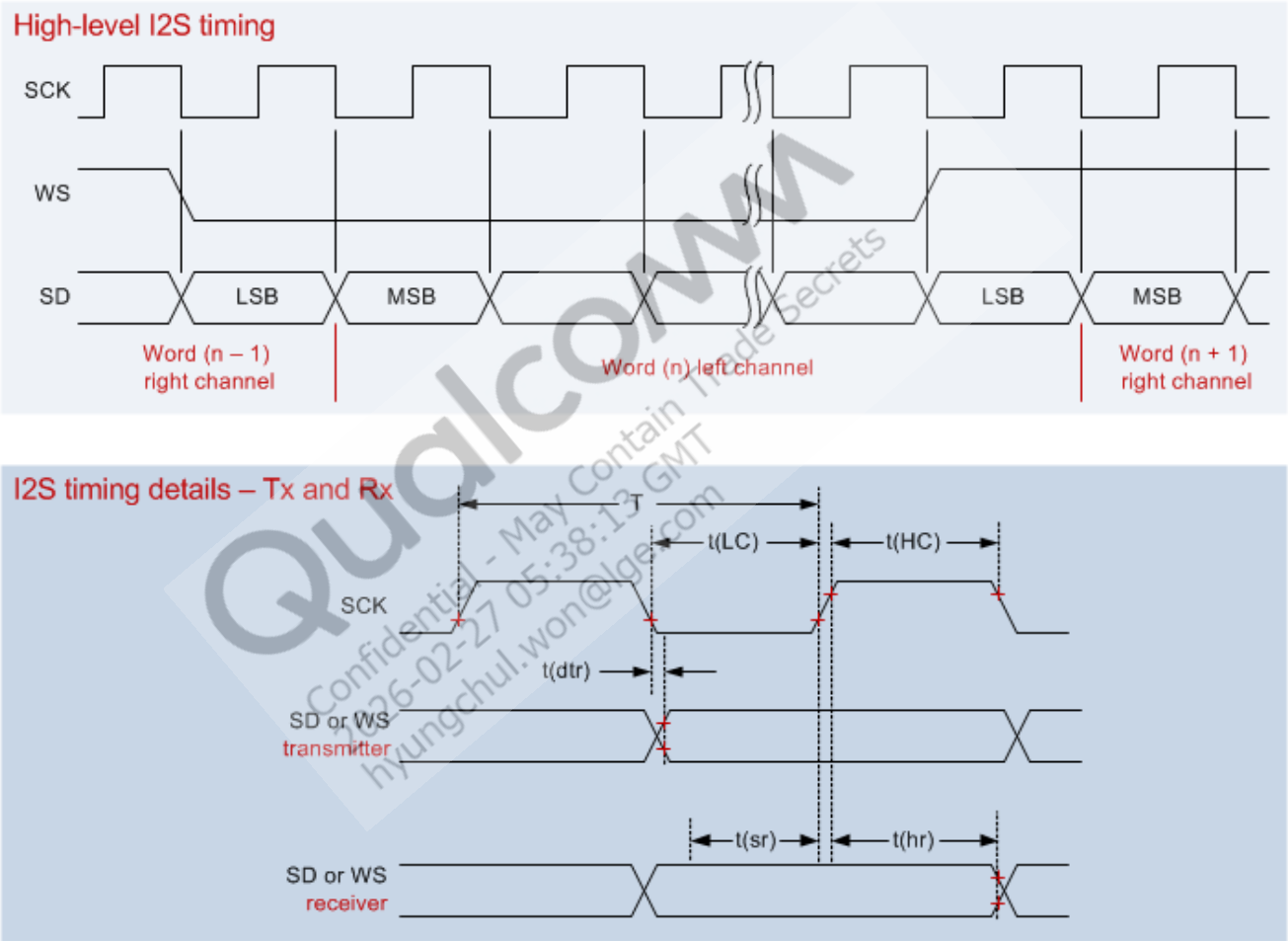

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Figure 3-12 I²S timing diagram

Table 3-22 I²S interface timing specifications

Parameter		Comments [a]	Min	Typ	Max	Unit
Using internal SCK						
Frequency			–	–	24.576	MHz
T	Clock period		40.69	–	–	ns

Table 3-22 I²S interface timing specifications (cont.)

Parameter		Comments [a]	Min	Typ	Max	Unit
t(HC)	Clock high		0.40 × T	–	0.60 × T	ns
t(LC)	Clock low		0.40 × T	–	0.60 × T	ns
t(sr)	SD and WS input setup time		0.20 × T	–	–	ns
t(hr)	SD and WS input hold time		1.5	–	–	ns
t(dtr)	SD and WS output delay		–	–	0.20 × T	ns

[a] Load capacitance is between 10 pF and 40 pF.

3.10.8 PCM/TDM interfaces

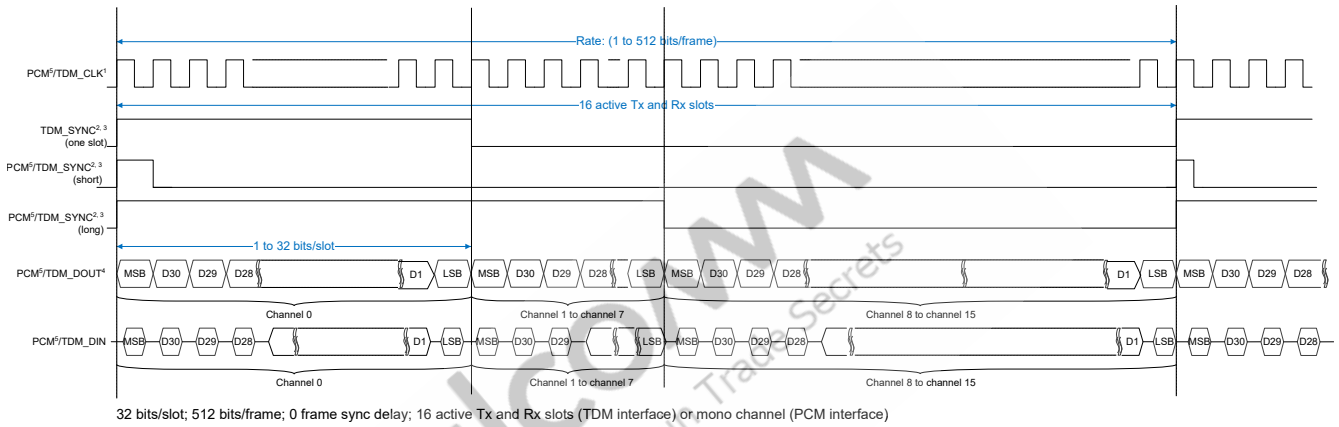

Figure 3-13 PCM/TDM audio format with different sync modes

NOTE

- Internal clock can also be inverted (180 degrees out of phase) relative to the external clock.
- Frame sync signal can also be inverted.
- Supports 0 to 2 cycle delays between the frame sync pulse edge and PCM_DOUT/DIN data.
- PCM data per slot can be smaller or equal to the slot size:
  - If data size < slot size, remaining data bits are padded with zeroes.
  - If data size > slot size, extra data bits will be ignored.
- PCM audio interface:
  - Supports only mono channel.
  - Does not support one-slot mode.
  - PCM_SYNC period is equivalent to 1 frame.

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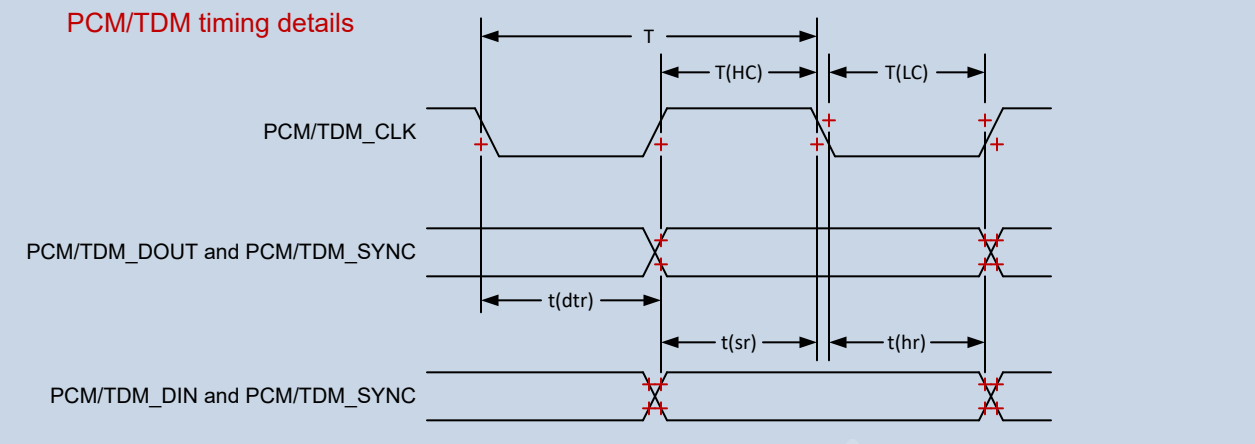

Figure 3-14 PCM/TDM timing diagram

Table 3-23 PCM/TDM interface timing parameters

Parameter [a]		Min	Max	Unit
Master mode				
Frequency		N/A	24.576	MHz
T	Clock period	40.69	N/A	ns
t(HC)	Clock high	0.45 × T	0.55 × T	ns
t(LC)	Clock low	0.45 × T	0.55 × T	ns
t(sr)	PCM/TDM_DIN and PCM/TDM_SYNC setup time	8.14	N/A	ns
t(hr)	PCM/TDM_DIN and PCM/TDM_SYNC hold time	1.5	N/A	ns
t(dtr)	PCM/TDM_DOUT and PCM/TDM_SYNC output delay	N/A	8.14	ns
Slave mode				
Frequency		N/A	24.576	MHz
T	Clock period	40.69	0.60 × T	ns
t(HC)	Clock high	0.45 × T	0.55 × T	ns
t(LC)	Clock low	0.45 × T	0.55 × T	ns
t(sr)	PCM/TDM_DIN and PCM/TDM_SYNC setup time	8.14	N/A	ns
t(hr)	PCM/TDM_DIN and PCM/TDM_SYNC hold time	1.5	N/A	ns
t(dtr)	PCM/TDM_DOUT and PCM/TDM_SYNC output delay	N/A	8.14	ns

[a] Load capacitance is between 10 pF to 40 pF.

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3.10.9 I³C interface

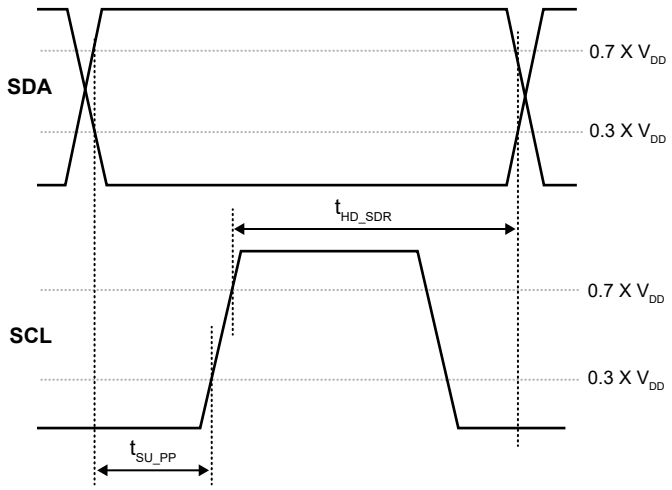

Figure 3-15 Push Pull mode master SDR mode timing diagram

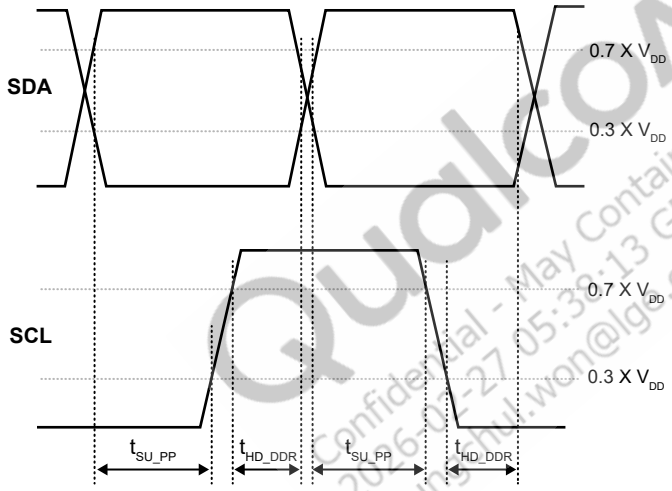

Figure 3-16 Push Pull mode master DDR mode timing diagram

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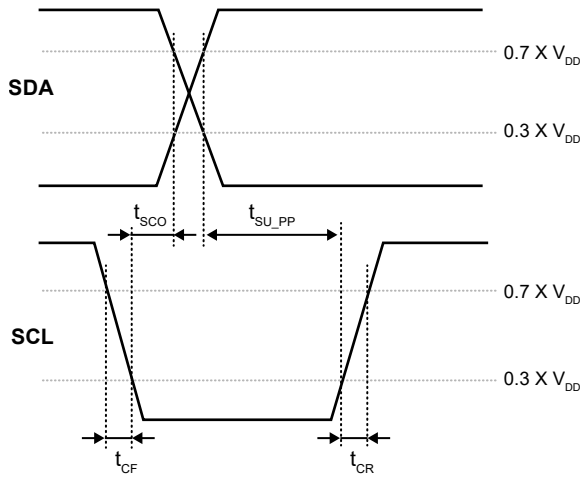

Figure 3-17 I³C slave out timing

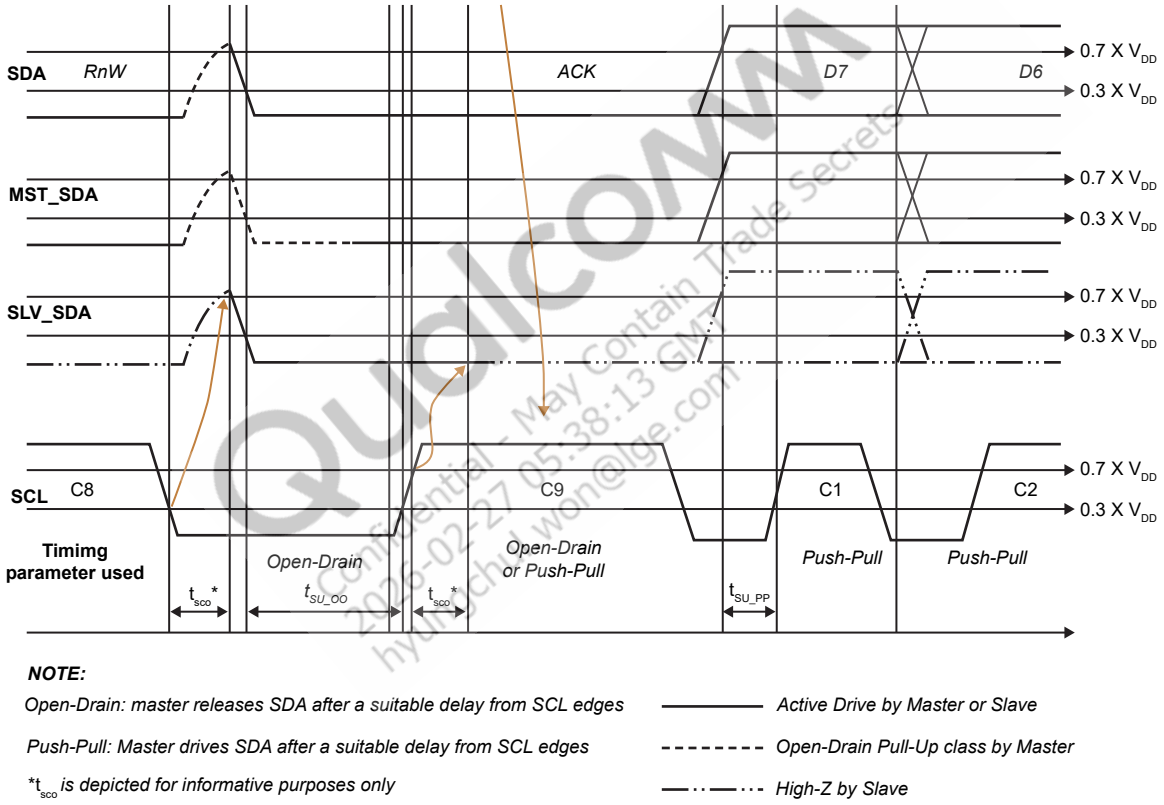

Figure 3-18 Open drain mode timing diagram

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Table 3-24 AC timing parameters: I³C master timing except instances of SSC se8 and se9

Symbol	Parameter	Min	Typ	Max	Units	Comments
fSCL	SCL clock frequency	0.01	12.5	12.9	MHz	–
tSU	SDA signal data setup time	13	–	–	ns	–
tHD	SDA signal data hold time	15	–	–	ns	–
tSCO	Clock IN to data out for slave	–	–	12	ns	Only supports slave mode

NOTE Master at 12.5 MHz for PP mode and 1 MHz for open drain mode.

3.10.10 Serial peripheral interface

The SW6100/SW6100 chipsets support SPI as a master only. Any one of the QUP ports can be configured as an SPI master.

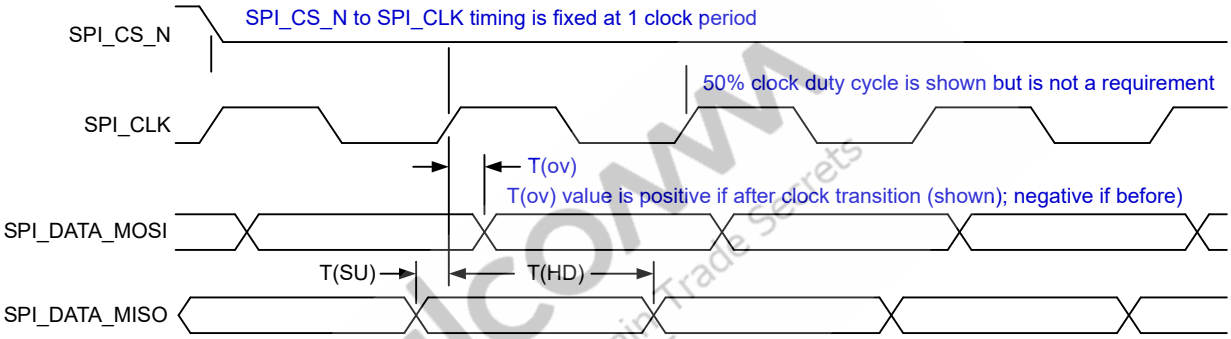

Figure 3-19 SPI master timing diagram

3.10.10.1 Master mode

Table 3-25 SPI master Tx modes of characterization

MODE index	CPOL	CPHA	SCLK edge for characterization (Full cycle timing)
MODE0	0	0	Fall edge
MODE1	0	1	Rise edge
MODE2	1	0	Rise edge
MODE3	1	1	Fall edge

Table 3-26 SPI master Rx modes of characterization

MODE index	CPOL	CPHA	SCLK edge for characterization (Full cycle timing)
MODE0	0	0	Fall edge
MODE1	0	1	Not supported
MODE2	1	0	Rise edge
MODE3	1	1	Not supported

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Table 3-27 SPI master timing (master at 50 MHz, 25 MHz)

Symbol	Parameter	Min	Max	Unit
Frequency	M_SPI_CLK at HV, TUR, NOM, SVS	–	50	MHz
	M_SPI_CLK at LSVS	–	25	
T_{OV}	M_OUTPUT_VALID	-5	+5	ns
T_{SU}	M_INPUT_SETUP	5	–	ns
T_{HD}	M_INPUT_HOLD	1	–	ns

3.10.10.2 Slave mode

Table 3-28 SPI slave timing high performance (slave at 50 MHz)

Symbol	Parameter	Min	Max	Unit
Frequency	S_SPI_CLK at HV, TUR, NOM	–	50	MHz
T_{OV}	S_OUTPUT_VALID	2	13	ns
T_{SU}	S_INPUT_SETUP	3	–	ns
T_{HD}	S_INPUT_HOLD	3	–	ns
T_{CS_en}	S_TRI_STATE_EN	–	14	ns
T_{CS_dis}	S_TRI_STATE_DIS	–	14	ns

Table 3-29 SPI slave timing (slave at 25 MHz, 12.5 MHz)

Symbol	Parameter	Min	Max	Unit
Frequency	S_SPI_CLK at SVS	–	25	MHz
	S_SPI_CLK at LSVS	–	12.5	
T_{OV}	S_OUTPUT_VALID	2	17	ns
T_{SU}	S_INPUT_SETUP	6	–	ns
T_{HD}	S_INPUT_HOLD	6	–	ns
T_{CS_en}	S_TRI_STATE_EN	2	17	ns
T_{CS_dis}	S_TRI_STATE_DIS	2	17	ns

NOTE For SPI timing characteristics, see *Serial Peripheral Interface Protocol Specification (80-VD509-1)*.

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3.10.11 SPMI interface

Timing specifications for master SCLK driver

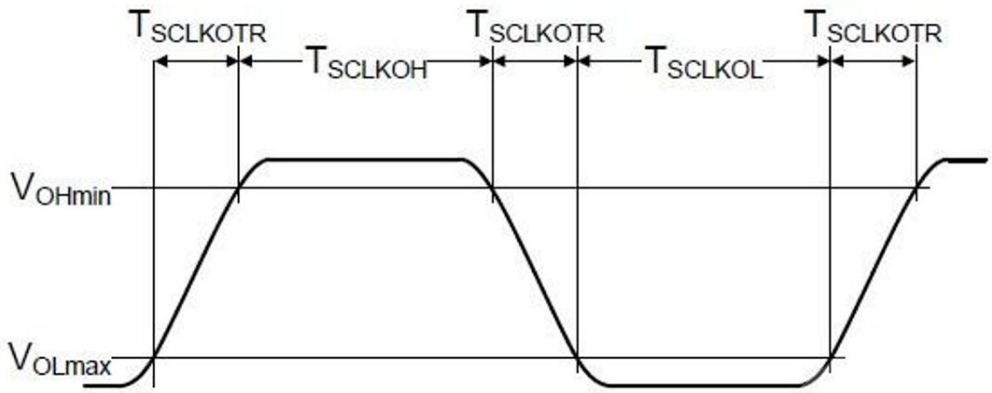

Figure 3-20 Clock driver output waveform

Table 3-30 Primary mode (eSPMI0) SCLK output timing characteristics (38.4 MHz)

Parameter	Description	Low speed device		High speed device		Unit
		Min	Max	Min	Max	
T_{sclkoh}	Clock output high	–	–	8.7	–	ns
T_{sclkol}	Clock output low	–	–	8.7	–	ns

Timing specifications for SDATA driver

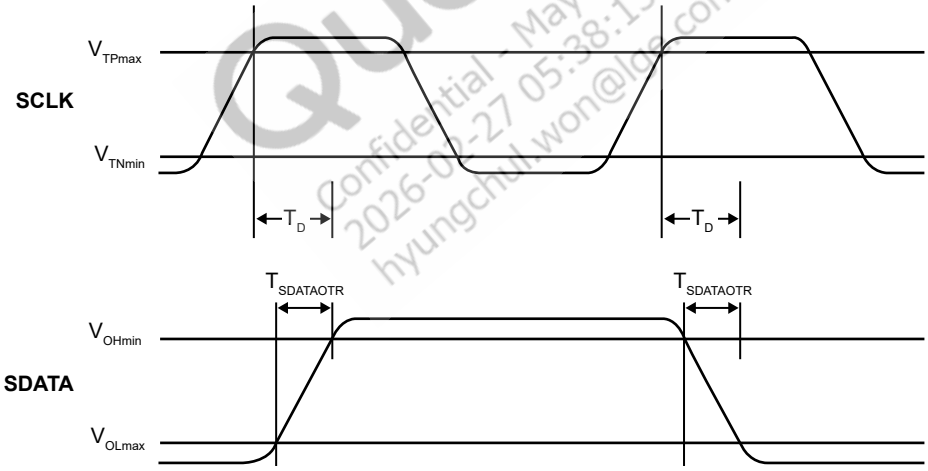

Figure 3-21 Bus active data transmission timing

Table 3-31 Primary mode (eSPMI0) SDATA output timing characteristics (38.4 MHz)

Parameter	Description	Low speed		High speed		Unit
		Min	Max	Min	Max	
T_d	Time for data valid from SCLK rising edge	–	–	-3.9	3.9	ns

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Timing specifications for SDATA receiver

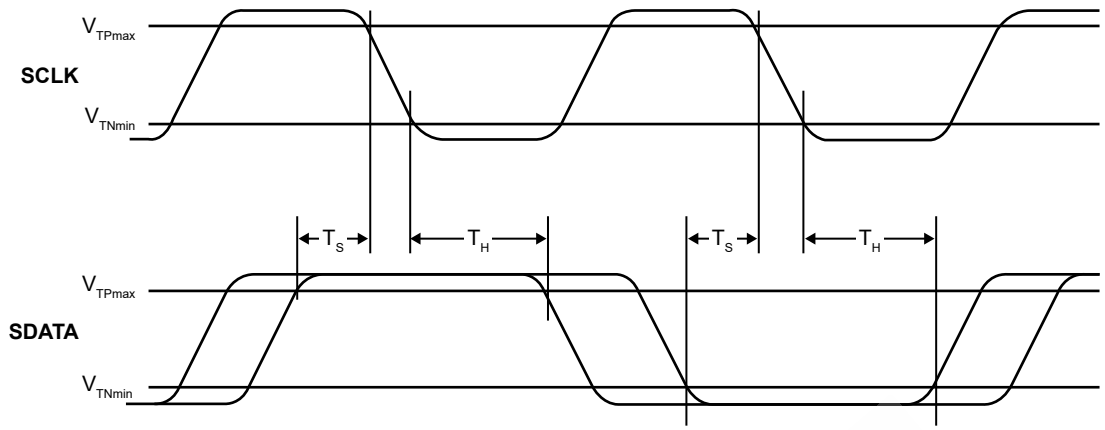
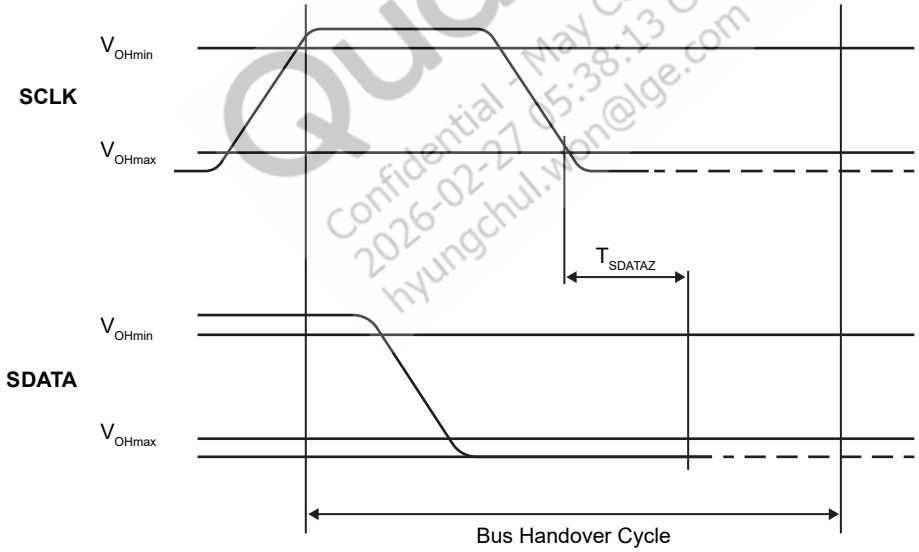


Figure 3-22 Bus active data receiver timing requirements

Table 3-32 Primary mode (eSPMI0) data receiver timing requirements (38.4 MHz)

Parameter	Description	Low speed		High speed		Unit
		Min	Max	Min	Max	
T_S	Data setup time	–	–	3	–	ns
T_H	Data hold time	–	–	3	–	ns

Timing specifications for SDATA release time



—— Signal driven – – – Signal not driven; pull down only

T_{SDATAZ} is measured from SCLK VOL level for the Master device driving SCLK and SDATA line.

Figure 3-23 Bus handover timing

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Table 3-33 Primary mode (eSPMI0) SDATA timing release parameters (38.4 MHz)

Parameter	Description	Low speed		High speed		Unit
		Min	Max	Min	Max	
T_{SDATAz}	Data drive release time	–	–	-8.3	–	ns

3.10.12 SWD interface

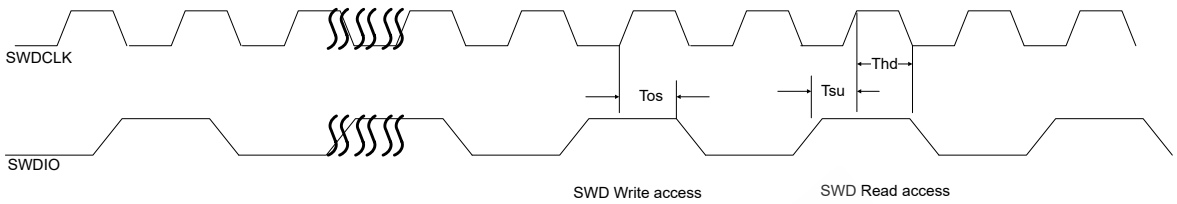

Figure 3-24 SWD write and read AC timing diagram

Table 3-34 SWD AC timing parameters

Parameter		Min	Max	Unit
T_{os}	SWDIO output skew to the rising edge of SWDCLK	-1	17.5	ns
T_{su}	Input setup time between SWDIO and the rising edge of SWDCLK	6.5	–	ns
T_{hd}	Input hold time between SWDIO and the rising edge of SWDCLK	6.5	–	ns

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3.11 Internal functions

Some internal functions require external interfaces to enable their operation. These include clock generation, modes and resets, and JTAG functions.

3.11.1 Clocks

Clocks that are specific to particular functions are addressed in the corresponding sections of this document. Others are specified here.

3.11.1.1 19.2 MHz CXO input

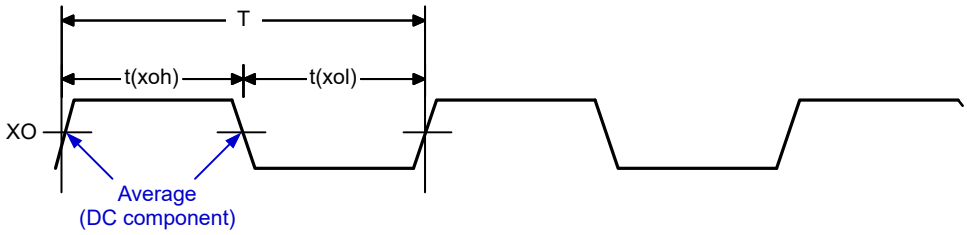

Figure 3-25 XO timing parameters

Table 3-35 CXO timing parameters

Parameter		Comments [a]	Min	Typ	Max	Unit
t(xoh)	XO logic high	–	22.6	–	29.5	ns
t(xol)	XO logic low	–	22.6	–	29.5	ns
T	XO clock period	–	–	52.083	–	ns
1/T	Frequency	19.2 MHz must be used	–	19.2	–	MHz

[a] For more information, see PMIC Data Sheets, *PW6100-0 Data Sheet (80-83960-1)* and *PW6100-1 Data Sheet (80-83961-1)*.

3.11.1.2 Sleep clock

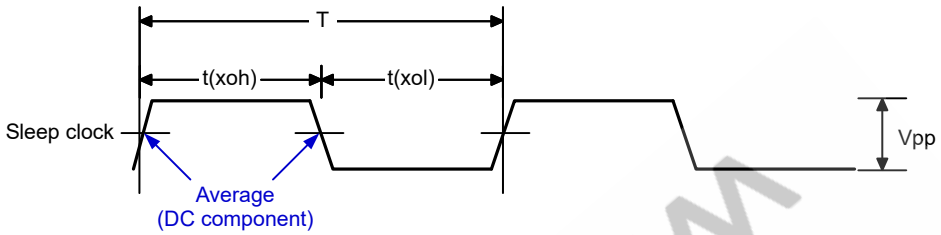

Figure 3-26 Sleep-clock timing parameters

Table 3-36 Sleep-clock timing parameters

Parameter		Comments	Min	Typ	Max	Unit
t(xoh)	Sleep-clock logic high	–	4.58	–	25.94	µs
t(xol)	Sleep-clock logic low	–	4.58	–	25.94	µs
T	Sleep-clock period	–	–	30.518	–	µs
F	Sleep-clock frequency	F = 1/T	–	32.768	–	kHz
Vpp	Peak-to-peak voltage	–	–	1.8	–	V

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3.11.2 Modes and resets

Mode and reset functions are basic digital I/Os that meet the performance specifications presented in Digital-logic characteristics.

3.11.3 JTAG

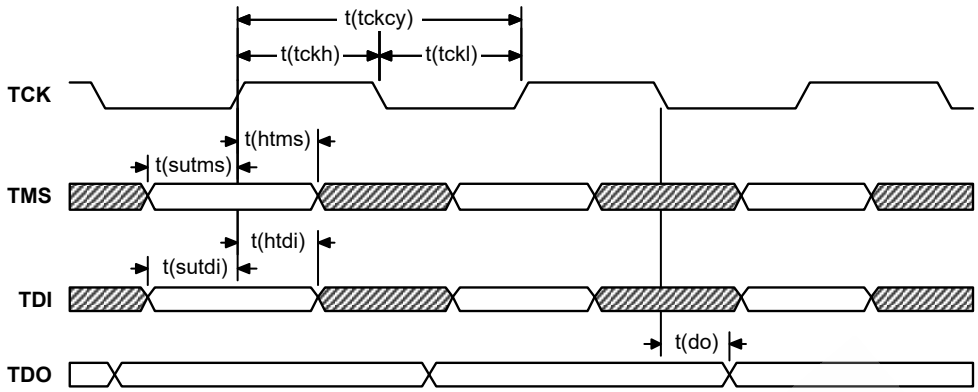

Figure 3-27 JTAG interface timing diagram

Table 3-37 JTAG interface timing characteristics

Parameter		Min	Typ	Max	Unit
t(tckcy)	TCK period	50	–	–	ns
t(tckh)	TCK pulse width high	20	–	–	ns
t(tckl)	TCK pulse width low	20	–	–	ns
t(sutms)	TMS input setup time	5	–	–	ns
t(htms)	TMS input hold time	20	–	–	ns
t(sutdi)	TDI input setup time	5	–	–	ns
t(htdi)	TDI input hold time	20	–	–	ns
t(do)	TDO data output delay	–	–	15	ns

NOTE JTAG clock (TCK) is tested up to 10 MHz.

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3.12 Power management interface

The digital I/Os must meet the logic-level requirements specified in Digital-logic characteristics.

3.12.1 RF and GRFC interfaces

Table 3-38 Supported RF and GRFC interfaces

Interfaces	Description
RF operating bands	▪ Defined by the RF transceiver SDR125 ▪ 5G NR sub-6 (FR1) SA and multimode 4G LTE
GRFC and RFFE interfaces	13 GRFC and 7 RFFE

3.12.2 System power management interface (SPMI)

Table 3-39 Supported SPMI standards and exceptions

Applicable standard	Feature exceptions
MIPI Alliance Specification for System Power Management Interface (SPMI) version 1.0	None

3.13 QMSPI specifications

Tx specifications

The Tx specifications are described in the table and figure below:

Table 3-40 Maximum load supported: 15 pf

Parameter		Comments	Min	Typ	Max	Unit
t_{HC}	Clock high	–	0.45 × T	–	0.55 × T	ns
t_{LC}	Clock low	–	0.45 × T	–	0.55 × T	ns
t_{oDLY}	Output delay	> 200 MHz	0.4	–	T/2-0.4	ns
		< 200 MHz	0.6	–	T/2-0.6	ns

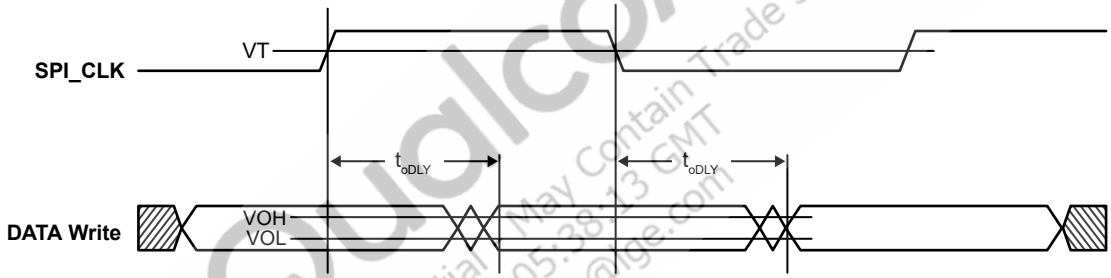

Figure 3-28 Tx specifications

Rx specifications

The Rx specifications are described in the table and figure below:

Table 3-41 Rx specifications(≥ 100 MHz frequency)

Parameter		Comments	Min	Max	Unit
TDQSQ	DQS to DQ skew	≥ 200 MHz	–	0.4	ns
		166 MHz	–	0.5	ns
		133 MHz	–	0.6	ns

Table 3-42 Rx specifications(≤ 100 MHz frequency)

Parameter		Min	Max	Unit
t_{Isu}	Data input setup	2	–	ns
t_{IH}	Data input hold	1	–	ns

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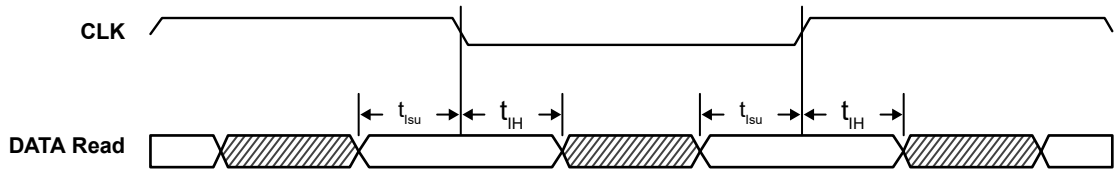

Figure 3-29 Rx specifications

NOTE QMSPI interface is tested only up to 200 MHz on the Qualcomm reference platform. If the clock frequency is higher than 200 MHz, contact QTI for support.

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4 Mechanical information

4.1 Device physical dimensions

The SW6100-1/SW6100P-1 is available in the FOPPSP646 that includes dedicated ground pins for improved grounding, mechanical strength, and thermal continuity. The FOPPSP646 has a 10.0 mm by 8.0 mm body, with a maximum height of 0.36 mm. Pin A1 is located by an indicator mark on the top of the package, and by the ball pattern when viewed from below. A simplified version of the FOPPSP 646 outline drawing is shown in the figure below.

NOTE Maximum height specified is exclusive of PoP memory.

NOTE Click the following link to download the *Package Outline Drawing, FOPPSP646, 10.0 × 8.0 × 0.36 mm, RBS32.5, M160, RFS24, PSPPB 201*:

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/NT90-38834-1

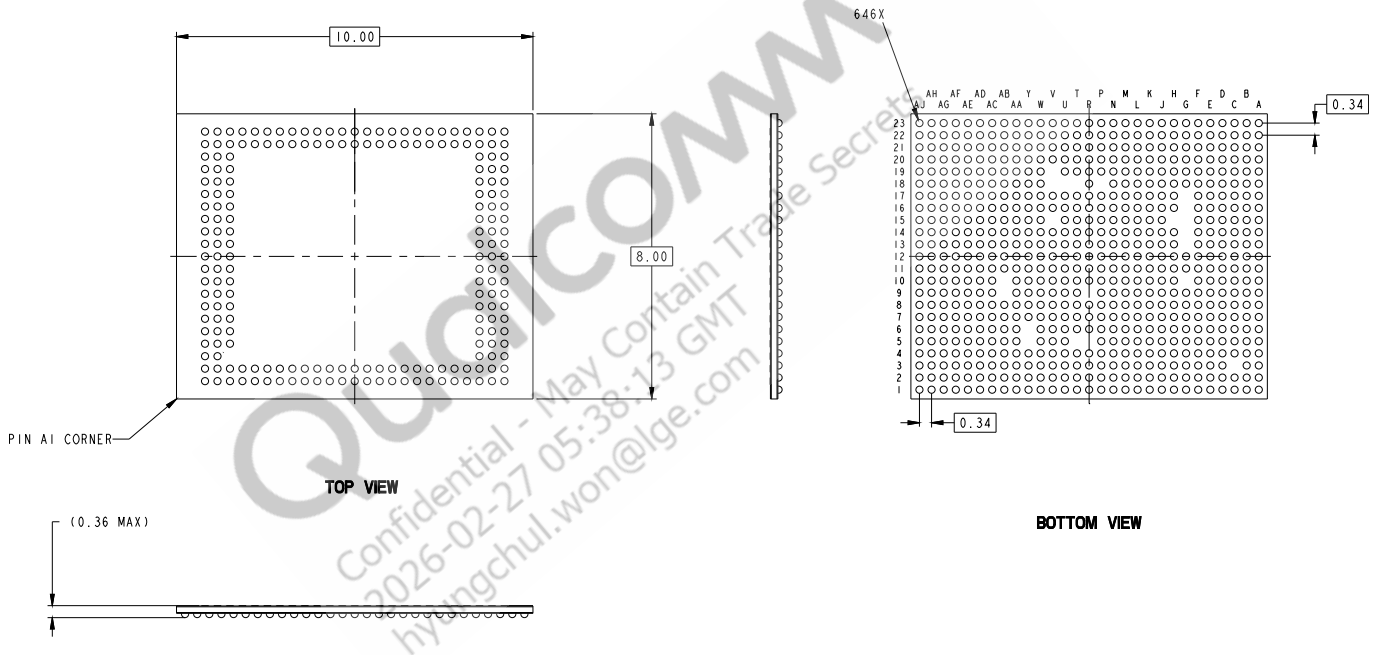

Figure 4-1 FOPPSP646 outline drawing

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4.2 Part marking

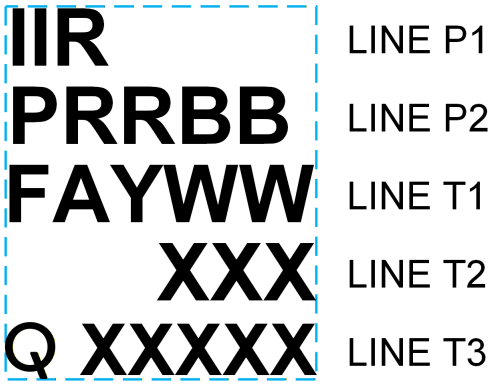

Figure 4-2 Device marking (top view, not to scale) – Final version

Table 4-1 Device marking line definitions

Line	Marking	Description
P1	IIR	Variant code. See Table 4-4.
P2	PRRBB	Device variant information ▪ See Table 4-4 for the assigned values.
T1	FAYWW	F = supply source code ▪ F = F (TSMC) A = assembly site code ▪ A = E (ASE, Taiwan) ▪ A = K (SPIL, Taiwan) Y = single/last digit of year WW = two-digit work week of year specified by Y
T2	XXX	traceability number
T3	Q	Pin1 or pin A1 indicator
	XXXXX	Additional trace information

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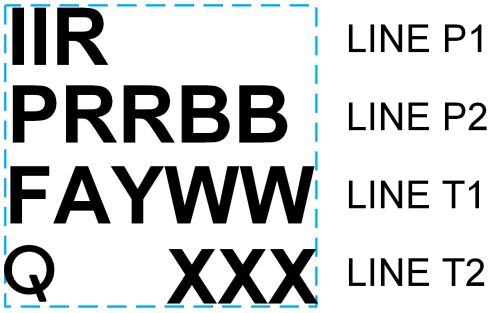

Figure 4-3 Device marking (top view, not to scale) – Earlier version

Table 4-2 Device marking line definitions

Line	Marking	Description
P1	IIR	Variant code. See Table 4-4.
P2	PRRBB	Device variant information ▪ See Table 4-4 for the assigned values.
T1	FAYWW	F = supply source code ▪ F = F (TSMC) A = assembly site code ▪ A = E (ASE, Taiwan) ▪ A = K (SPIL, Taiwan) Y = single/last digit of year WW = two-digit work week of year specified by Y
T2	Q	Pin1 or pin A1 indicator
	XXX	traceability number

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4.3 Device ordering information

The Oracle short description is used to order QTI products, and is present on both the customer label and this document. The short description includes the product name, configuration code, package type, product revision code, source code, and feature code/program ID of the part.

This device can be ordered using the identification code shown in the following table.

Table 4-3 Example device identification code

Device ID code	AAA-AAAA	-P	-TTTTTT	NNNN	A	+FF	-EE	-RR	-S	-BB or-PID [a]
Symbol definition	**Product name**	**Configuration code**	**Package type**	**Number of pins**	**Package variable**	**Additional package information**	**Shipping package**	**Product revision**	**Source code**	**Feature code**
Example 1	SW-6100	-1	-FOPPSP	646			-MT	-K0	-0	-AA
Example 2	SW-6100P	-1	-FOPPSP	646			-MT	-L0	-0	-AA

[a] The feature code (BB) and the program ID (PID) are mutually exclusive. A product may have one of them or none of them, but it will never have both. If there is no feature code/program ID, this field is blank, and the Oracle short description ends after the source configuration code (S).

For example:

- Example 1: SW-6100-1-FOPPSP646-MT-K0-0-AA
- Example 2: SW-6100P-1-FOPPSP646-MT-L0-0-AA

NOTE The shipping package is either TR (tape and reel) or MT (matrix tray).

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4.4 Device identification for each sample type

Table 4-4 Device identification details

Device	Sample type	Item IdentifieR (IIR)	Variant (PRR-BB) P= product configuration code RR = product revision code BB= feature code (if applicable) [a]	Hardware revision number (JTAG_ID – see Table 4-1)	Source configuration code (S) [b]	Comments	Sample date
SW6100	ES	7K0	1K0-AA	0x0 030A 0E1	0	SW6100 (1K0-0-AA), FOPPSP646 package, 1 × Arm Cortex-A78, 4 × Arm Cortex-A55, full scope RDL configuration with modem	06/13/2025
SW6100P	ES	7L0	1L0-AA	0x0 030B 0E1	0	SW6100P (1L0-0-AA), FOPPSP646 package, 1 × Arm Cortex-A78, 4 × Arm Cortex-A55, full scope RDL configuration without modem and SDR125	06/13/2025
SW6100	CS [c d] CS date code are as follows: ▪ ASE, Taiwan: 601 ▪ SPIL, Taiwan: TBD	7K0	1K0-AA	0x0 030A 0E1	0	SW6100 (1K0-0-AA), FOPPSP646 package, 1 × Arm Cortex-A78, 4 × Arm Cortex-A55, full scope RDL configuration with modem	2/12/2026
SW6100P	CS [cd] CS date code are as follows: ▪ ASE, Taiwan: 601 ▪ SPIL, Taiwan: TBD	7L0	1L0-AA	0x0 030B 0E1	0	SW6100P (1L0-0-AA), FOPPSP646 package, 1 × Arm Cortex-A78, 4 × Arm Cortex-A55, full scope RDL configuration without modem and SDR125	2/12/2026

[a] BB is the feature code that identifies an IC's specific feature set, which distinguishes it from other versions or variants. Feature sets are detailed in the Comments column.
[b] S is the source configuration code that identifies all of the qualified die fabrication-source combinations available when the particular sample type was shipped. S values are defined in Table 4-5.
[c] All devices with YWW date code or later are CS quality. However, devices with earlier date codes may also be CS quality. Contact your customer service representative for CS part marking with date codes prior to YWW.
[d] ES material may be shipped post CS sample date upon customer request.

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Table 4-5 Source configuration code

S value	F value = F	A value = E	A value = K
0	TSMC	ASE, Taiwan	SPIL, Taiwan
Other columns and rows will be added in future revisions of this document, if needed.			

4.5 Thermal characteristics

Rather than providing thermal resistance values θ_{JC} and θ_{JA}, validated thermal package models are provided through TBD. Designers can extract thermal resistance values by conducting their own thermal simulations.

NOTE Click the following links to download the thermal models for SW6100-1/SW6100P-1 chipsets:

- *SW6100-1 Package Thermal Model Icepak (HS11-83951-5HW)*
- *SW6100-1 Package Thermal Model FloTHERM (HS11-83951-6HW)*

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5 Carrier, storage, and handling information

5.1 Manufacturers date code

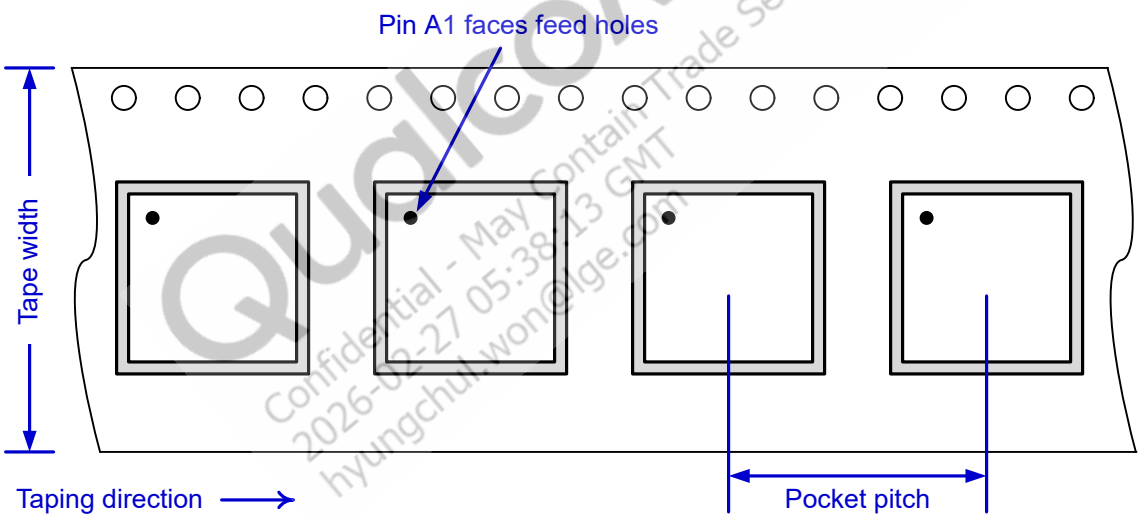
QTI's manufacturing date code has no impact on the usage, quality, or reliability of products due to advancements in our technology, materials, manufacturing methods, and logistics. Any material shipped to customers is subject to QTI standard terms and conditions.

5.2 Carrier

5.2.1 Tape and reel information

All QTI tape carrier systems conform to EIA-481 standards.

A simplified sketch of the SW6100-1/SW6100P-1 tape carrier is shown in the following figure, including the proper part orientation, maximum number of devices per reel, and key dimensions.



Tape feed:	Single	Reel diameter:	330 mm	Tape width:	24 mm
Units per reel:	4000	Hub diameter:	178 mm	Pocket pitch:	12 mm

Figure 5-1 Carrier tape drawing with part orientation

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Tape-handling recommendations are shown in the following figure.

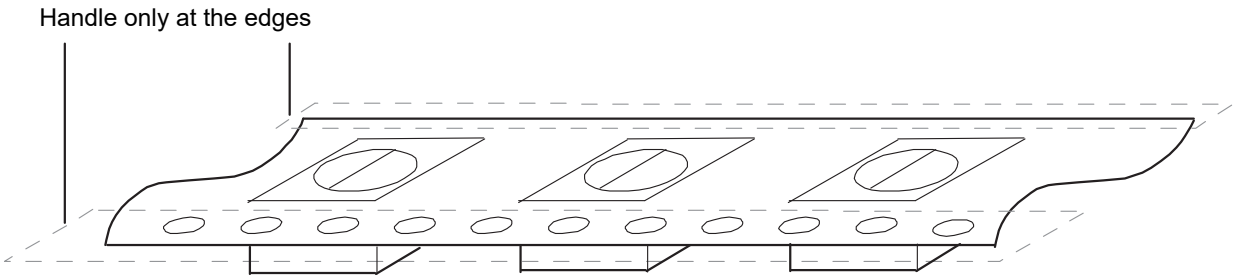

Figure 5-2 Tape handling

5.2.2 Matrix tray information

All QTI matrix tray carriers confirm to JEDEC standards.

The device pin 1 is oriented to the chamfered corner of the matrix tray.

Each tray of the SW6100-1/SW6100P-1 contains up to 180 devices. Production orders of the SW6100-1/SW6100P-1 that are shipped in matrix tray carriers will be in [10 + 1] tray stacks of [1800] units. The stacking configuration and quantity for sample orders will vary.

See the following figure for matrix-tray key attributes and dimensions.

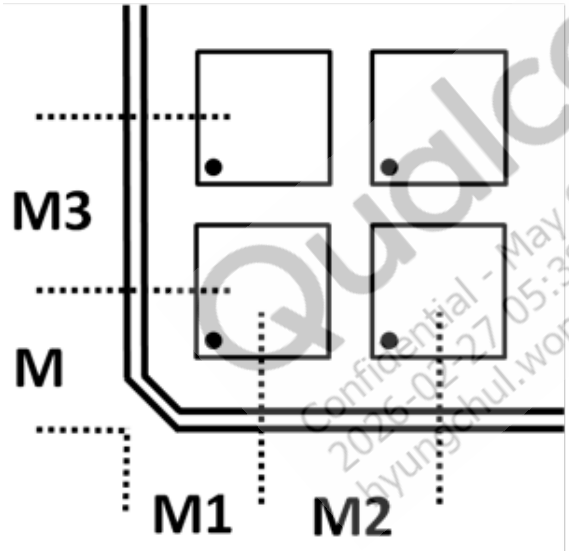

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Figure 5-3 Matrix-tray key attributes and dimensions

Table 5-1 Matrix-tray key attributes and dimensions

Key dimensions	
Array	10 × 21 (210)
M	10.35 mm
M1	11.00 mm
M2	14.65 mm
M3	12.80 mm

5.3 Device moisture sensitivity level

Plastic-encapsulated surface mount packages are susceptible to damage induced by absorbed moisture and high temperature. A package's moisture sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device. All pertinent MSL ratings are summarized in the following table.

Table 5-2 MSL ratings summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	≤ 30°C/85% RH
2	1 year	≤ 30°C/60% RH
2a	4 weeks	≤ 30°C/60% RH
3	168 hours	≤ 30°C/60% RH
4	72 hours	≤ 30°C/60% RH
5	48 hours	≤ 30°C/60% RH
5a	24 hours	≤ 30°C/60% RH
6	Mandatory baking before use. After baking, must be reflowed within the time limit specified on the label.	≤ 30°C/60% RH

QTI follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivity qualification. The SW6100-1/SW6100P-1 devices are classified as MSL 3; the qualification temperature was 260 +0/-5°C. This qualification temperature (260 +0/-5°C.) should not be confused with the peak temperature within the recommended solder reflow profile.

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5.4 Storage

5.4.1 Bagged storage conditions

SW6100-1/SW6100P-1 devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, antistatic bags. See the *IC Products Packing Method document* (80-VK055-1) for the expected shelf life.

5.4.2 Out-of-bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB. It is defined by the device MSL rating, as described in Device moisture sensitivity level.

5.5 Handling

Tape handling was described in Tape and reel information. Other (IC-specific) handling guidelines are presented in the following subsections.

5.5.1 Baking

It is not necessary to bake the SW6100-1/SW6100P-1 if the conditions specified in Bagged storage conditions and Out-of-bag duration have not been exceeded.

It is necessary to bake the SW6100-1/SW6100P-1 if any condition specified in Bagged storage conditions or Out-of-bag duration has been exceeded. The baking conditions are specified on the moisture-sensitive caution label attached to each bag; see the *IC Products Packing Method* (80-VK055-1) document for details.

CAUTION: If baking is required, the devices must be transferred into trays that can be baked to at least 125°C. Devices should not be baked in tape and reel carriers at any temperature.

5.5.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

QTI products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, Protection of Electrical and Electronic Parts, Assemblies, and Equipment.

5.6 Bar code label and packing for shipment

See the *IC Products Packing Method* (80-VK055-1) document for all packing-related information, including bar code label details.

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6 PCB mounting guidelines

6.1 RoHS compliance

The device complies with the requirements of the EU RoHS directive. Its SnAgCu solder balls use SAC125/Ni composition. A product material declaration (PMD) that provides RoHS and other product environmental governance information is published when the data is available.

6.2 SMT assembly guidelines

For recommendations on SMT process development, see the *SMT Assembly Guidelines* (SM80-P0982-1).

NOTE Click the following link to download the *SMT Assembly Guidelines (SM80-P0982-1)* from the CreatePoint webiste.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/SM80-P0982-1

After successfully logging in, the document is downloaded.

NOTE Make this document a favorite to be notified of any changes.

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7 Part reliability

This section will be updated in the future revisions of the document.

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8 Samples and known issues

8.1 Sample testing

See Table 4-4 for device identification details for all available samples.

8.1.1 Engineering samples (ES)

These devices undergo limited testing and sometimes have significant feature limitations. They are suitable to assist with PCB development, to conduct board-level electrical evaluation tests, and to explore manufacturing considerations. Engineering samples should not be used for product-level qualification.

8.1.2 Commercial samples (CS)

These devices undergo full production-level testing, and meet the specifications and features described in the data sheet, except as otherwise noted in this document. They have passed device-level qualification. Commercial samples are suitable for performance testing, and also for product-level production and qualification.

8.2 Compatible software releases

Each sample type is for use with SW6100.LAW.1.0 software versions.

Table 8-1 Software compatibility for each sample type (PRR value)

SW6100 variant	PRR	Compatible software
SW6100-1/SW6100P-1	1K0-AA	SW6100.LAW.1.0
	1L0-AA	

8.3 Known issues

There are no known issues for SW6100-1/SW6100P-1.

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9 Revision history

Bars appearing in the margin (as shown here) indicate where technical changes have occurred for this revision. The following table lists the technical content changes for all revisions.

Revision	Date	Description
AA	January 2025	Initial release
AB	February 2025	■ Section 2.2.2 *Pin descriptions-bottom*: Updated
AC	June 2023	■ Section 4 *Mechanical information*: Added the following subsections: ▫ Section 4.2 *Part marking* ▫ Section 4.3 *Device ordering information* ■ Section 3 *Electrical specification* ▫ Table 3-1 *Absolute maximum ratings*: Added ▫ Table 3-2 *Operating conditions*: Added ■ Section 5 *Carrier, storage and handling information: Added* ■ Section 6 *PCB mounting guidelines*: Added ■ Section 7 *Part reliability*: Added ■ Section 8 *Samples and known issues*: Added
AD	February 2026	■ Updated the following figures: ▫ Figure 1-1 SW6100-1 functional block diagram and example application ▫ Figure 1-2 SW6100P-1 functional block diagram and example application ■ Added the following sections: ▫ Section 1.2 *Device variants* ▫ Section 3.4 *DC power characteristics* ▫ Section 3.6 *Digital-logic characteristics* ▫ Section 3.7 *Timing characteristics* ▫ Section 3.8 *Memory support* ▫ Section 3.9 *Multimedia* ▫ Section 3.10 *Connectivity* ▫ Section 3.11 *Internal functions* ▫ Section 3.12 *Power management interface* ▫ Section 5.1 *Manufacturers date code* ▫ Section 8.2 *Compatible software releases* ■ Updated the following sections: ▫ Section 2.2.2 *Pin descriptions – bottom*: Updated pin descriptions for P6, AG4, G9, AH4, B8, and D7 ▫ Section 8.3 *Known issues*

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